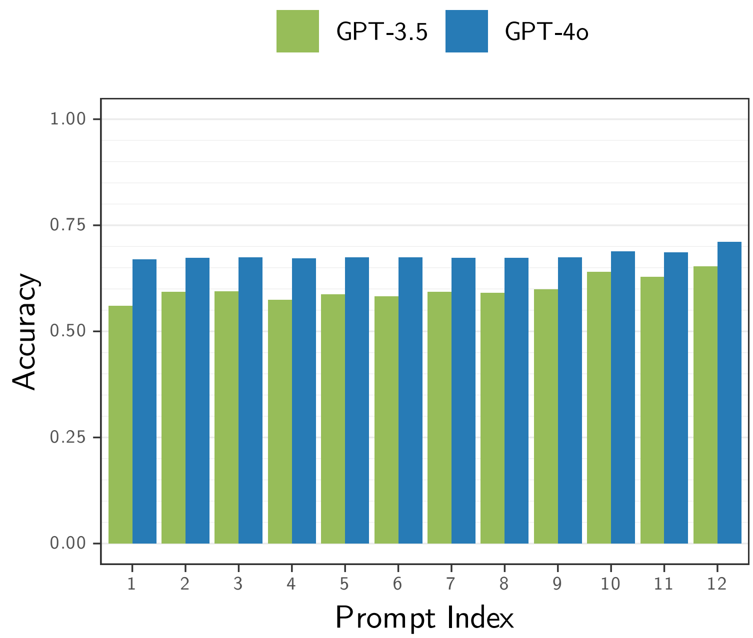
Results

Aug 12, 2024

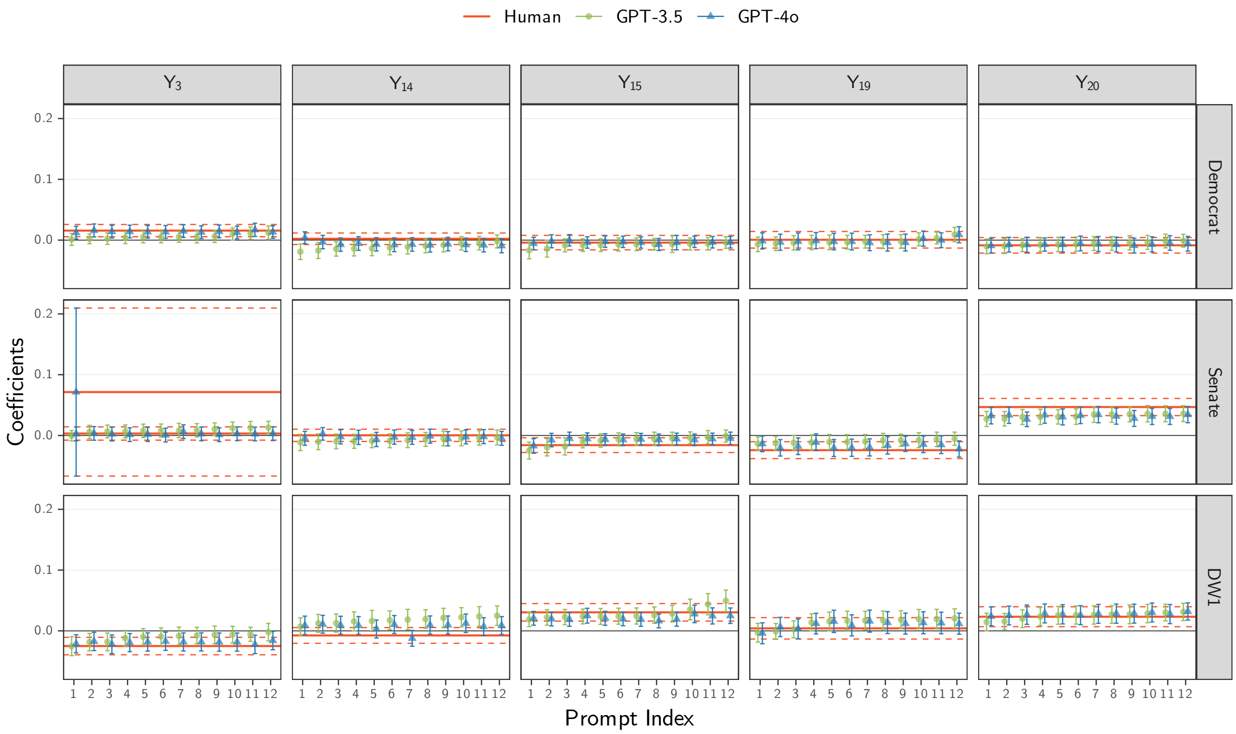
Figure 1: Accuracy of Topic Predictions vs. Prompt.



Note: Accuracy= $\frac{1}{10{,}000}\sum_{i=1}^{10{,}000} 1\{Y_{m,p}^{\text{LLM}} = Y^{\text{Human}}\}^{(i)}$

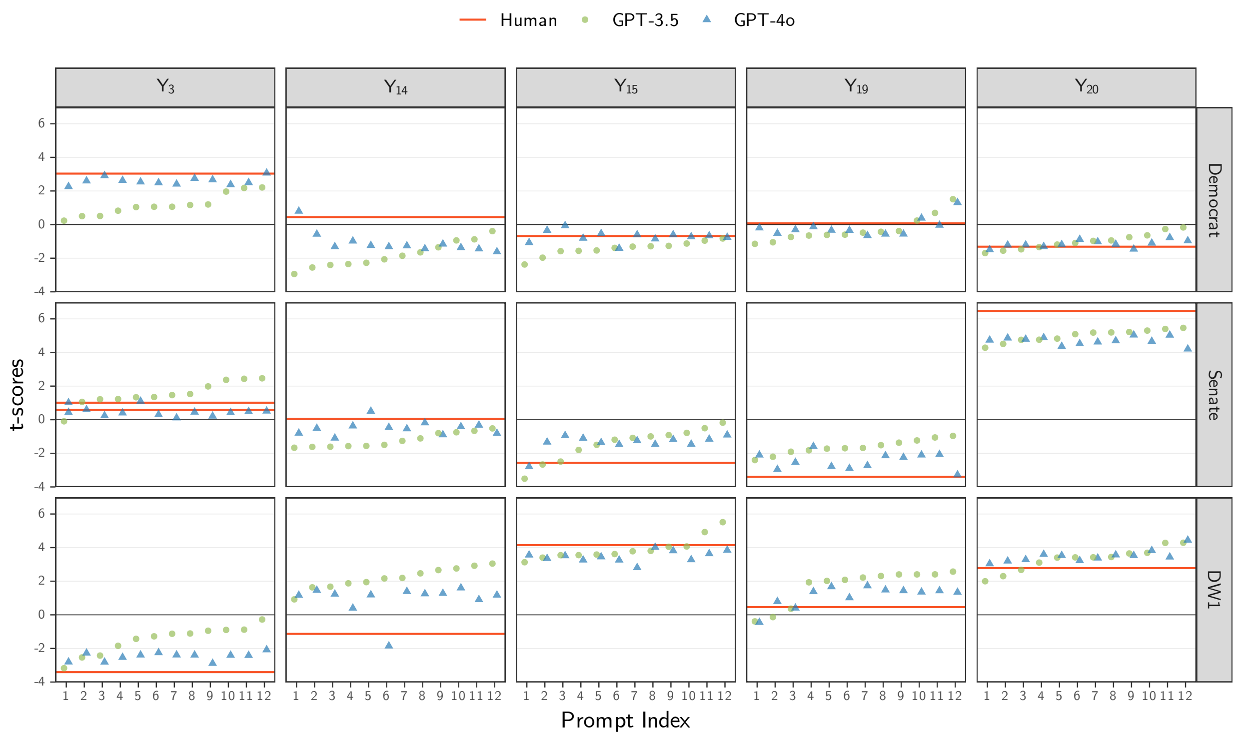
Proxy on the LHS

Figure 2: β_1 vs. Prompt. Proxy on the LHS.



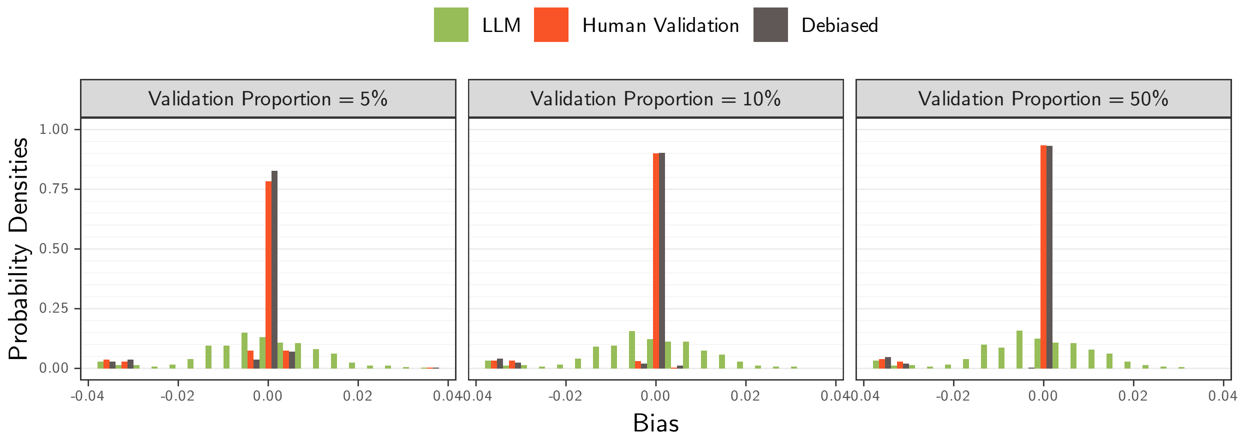
Note: The coefficients were obtained by fitting the model $Y_t = \beta_0 + \beta_1 V$ over 10,000 bills, with 95% confidence intervals computed using robust standard errors. Here, Y_t denotes the indicator for topic t, where $Y_t = 1\{Y = t\}$ for $t \in \mathcal{T}$ and $Y_{\text{Other}} = 1\{Y \notin \mathcal{T}\}$, with $\mathcal{T} = \{3, 14, 15, 19, 20\}$ is the 5 most-common human-labeled topics in the bills data. The variable V takes one of the following bills variables: Democrat, Senate, or DW1, where $extDemocrat = 1\{\text{Party} = \text{Democrat}\}$, $\text{Senate} = 1\{\text{Chamber} = \text{Senate}\}$, and DW1 is the imputed DW1 score for the bill sponsor. The regressions based on human-labeled topics (red) were estimated with $Y_t^{\text{Human}} = 1\{Y^{\text{Human}} = t\}$, while the LLM-based regressions (GPT-3.5 in green, circles; and GPT-4o in blue, triangles) were estimated with $Y_{t,m,p}^{\text{LLM}} = 1\{Y_{m,p}^{\text{LLM}} = t\}$, where $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$ denotes the LLM model and $p \in \{1, \ldots, 12\}$ denotes the prompt modifications. In each subplot, the indices for the prompts reflect the order chosen based on sorting the estimated values of the GPT-3.5 model in ascending order. The corresponding estimates for GPT-4o are aligned according to this order.

Figure 3: t-scores of β_1 vs. Prompt. Proxy on the LHS.



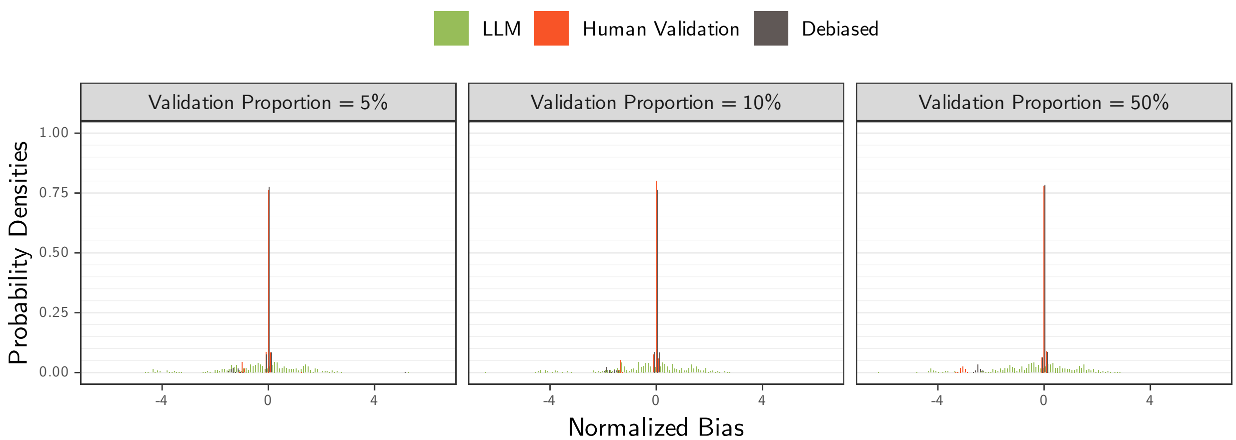
Note: The t-scores were obtained by fitting the model $Y_t = \beta_0 + \beta_1 V$ over 10,000 bills, with 95% confidence intervals computed using robust standard errors. Here, Y_t denotes the indicator for topic t, where $Y_t = 1\{Y = t\}$ for $t \in \mathcal{T}$ and $Y_{\text{Other}} = 1\{Y \notin \mathcal{T}\}$, with $\mathcal{T} = \{3, 14, 15, 19, 20\}$ is the 5 most-common human-labeled topics in the bills data. The variable V takes one of the following bills variables: Democrat, Senate, or DW1, where $extDemocrat = 1\{\text{Party} = \text{Democrat}\}$, Senate $= 1\{\text{Chamber} = \text{Senate}\}$, and DW1 is the imputed DW1 score for the bill sponsor. The regressions based on human-labeled topics (red) were estimated with $Y_t^{\text{Human}} = 1\{Y^{\text{Human}} = t\}$, while the LLM-based regressions (GPT-3.5 in green, circles; and GPT-4o in blue, triangles) were estimated with $Y_{t,m,p}^{\text{LLM}} = 1\{Y_{m,p}^{\text{LLM}} = t\}$, where $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$ denotes the LLM model and $p \in \{1, \ldots, 12\}$ denotes the prompt modifications. In each subplot, the indices for the prompts reflect the order chosen based on sorting the estimated values of the GPT-3.5 model in ascending order. The corresponding estimates for GPT-4o are aligned according to this order.

Figure 4: Bias Density of β_1 by Proportion of Validation Samples. Proxy on the LHS.



Note: The bias estimates were computed as the difference between the coefficient estimates from $N = 1000$ simulation runs, $\bar{\beta}_{t,V,\eta,m,p} = \sum_{i=1}^{N} \beta^{(i)}_{t,V,\eta,m,p}/N$, and the human-based coefficient from the entire set of 10,000 bills, $\beta^{\star}_{t,V}$, using a regression model of the form $Y_t = \beta_0 + \beta_1 V$, for topic $t \in \{3, 14, 15, 19, 20, \text{Other}\}$, variable $V \in \{\text{Democrat}, \text{Senate}, \text{DW1}\}$, validation proportion $\eta \in \{5\%, 10\%, 25\%, 50\%\}$, LLM model $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$, and prompt $p \in \{1, \ldots, 12\}$. For the LLM-based models (green), a random sample of 5,000 bills drawn from the original 10,000 was used, with LLM-based labels, $Y^{\text{LLM}}_{t,m,p} = 1\{Y^{\text{LLM}}_{m,p} = t\}$. For the other two models—human-based (red) and debiased-based (brown)—the 5,000 sample were split into two subsets: validation and primary, with the size of the validation set determined by the validation proportion, η. The results for $\eta = 25\%$ were excluded from the figures for clarity. In the human-based model, only the human labels in the validation set were used, $Y^{\text{Human}}_t = 1\{Y^{\text{Human}} = t\}$, whereas the debiased estimates use both the validation and primary sets, with Y^{Human}_t from validation set used to estimate measurement errors and $Y^{\text{LLM}}_{t,m,p}$ from the primary set used to obtain the coefficient estimates used to compute the reported bias.

Figure 5: Normalized Bias Density of β_1 by Proportion of Validation Samples. Proxy on the LHS.



Note: The normalized bias estimates were computed as the normalized difference between the coefficient estimates from $N =$ 1000 simulation runs, $\bar{\beta}_{t,V,\eta,m,p}$, and the human-based coefficient from the entire set of 10,000 bills, $\beta^{\star}_{t,V}$, divided by the sample standard deviation, $s_{t,V,\eta,m,p}$, where $s^2_{t,V,\eta,m,p} = \sum_{i=1}^{N}(\beta^{(i)}_{t,V,\eta,m,p} - \bar{\beta}_{t,V,\eta,m,p})^2/(N-1)$ and all variables are as defined previously.

Figure 6: Bias Density of β_1 by Model and Proportion of Validation Samples. Proxy on the LHS.

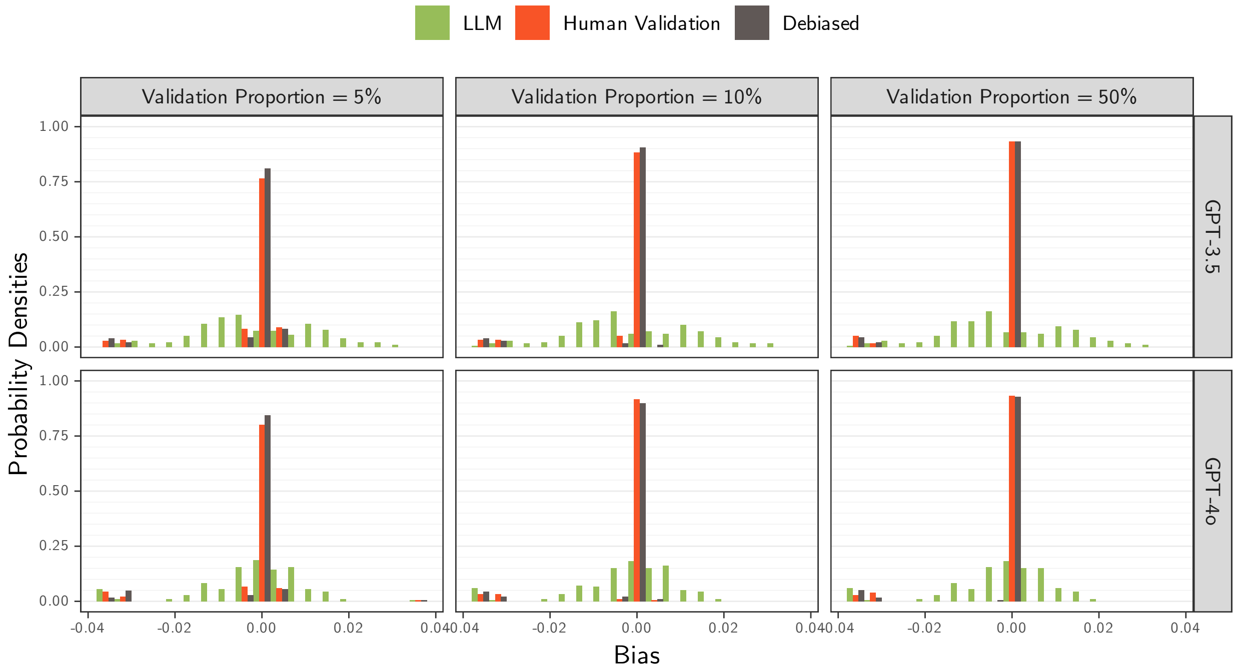


Figure 7: Normalized Bias Density of β_1 by Model and Proportion of Validation Samples. Proxy on the LHS.

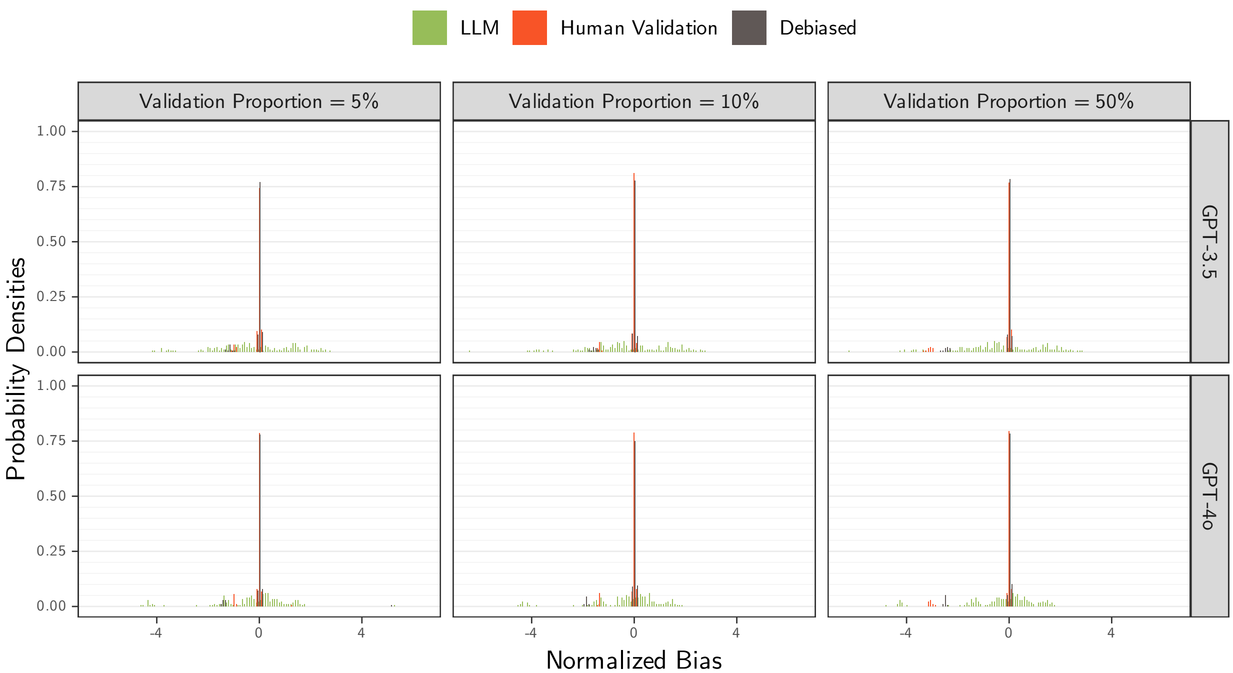


Figure 8: MSE Empirical CDF of β_1 by Proportion of Validation Samples. Proxy on the LHS.

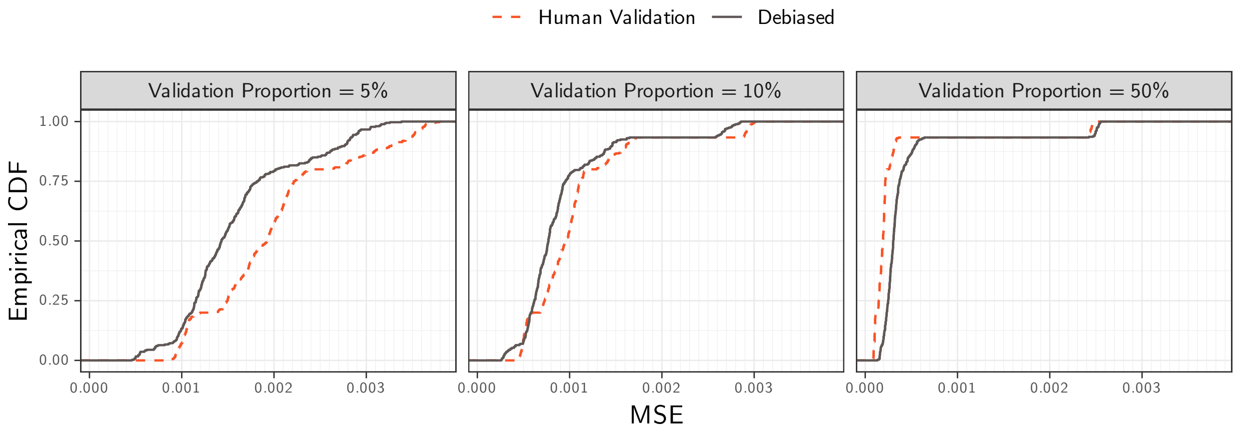


Figure 9: MSE Empirical CDF of β_1 by Model and Proportion of Validation Samples. Proxy on the LHS.

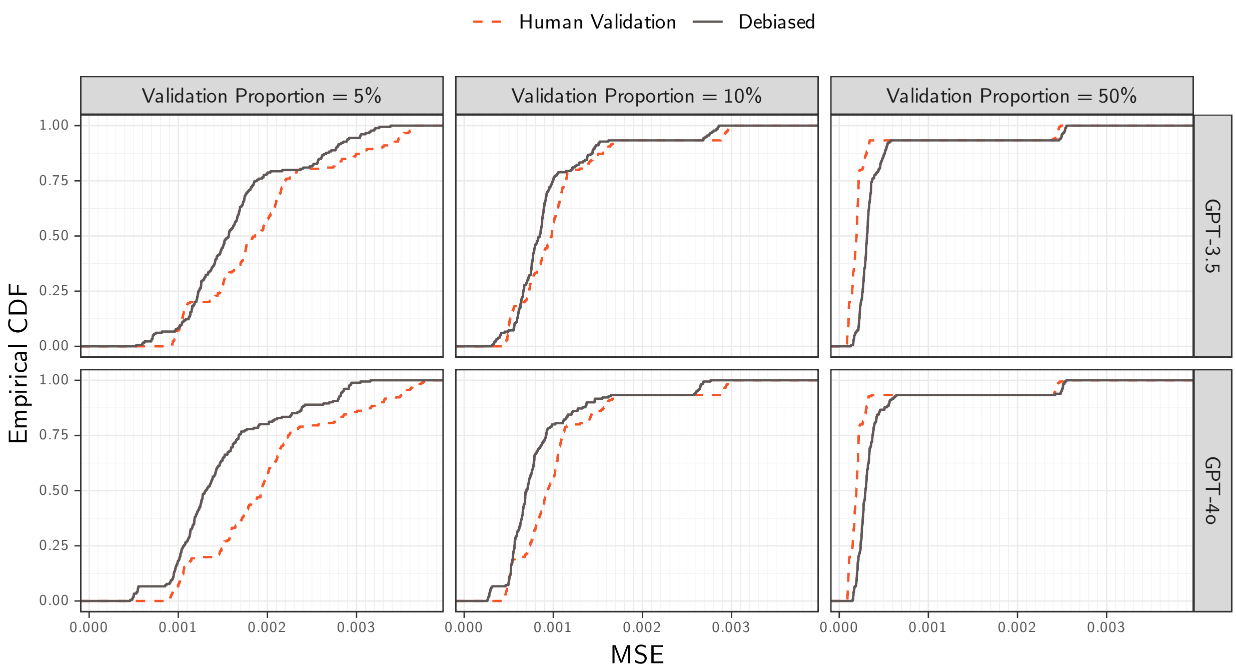


Table 1: Bias Summary Statistics of β_1 by Proportion of Validation Samples. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	-0.001	0.013	-0.001	-0.029	0.019
5%	Human Validation	-0.002	0.009	0.000	-0.033	0.002
5%	Debiased	-0.002	0.009	0.000	-0.033	0.002
10%	LLM	-0.001	0.013	-0.001	-0.029	0.018
10%	Human Validation	-0.002	0.009	0.000	-0.034	0.001
10%	Debiased	-0.002	0.009	0.000	-0.033	0.001
25%	LLM	-0.001	0.013	-0.001	-0.029	0.018
25%	Human Validation	-0.002	0.009	0.000	-0.034	0.001
25%	Debiased	-0.002	0.009	0.000	-0.034	0.001
50%	LLM	-0.001	0.013	-0.001	-0.029	0.019
50%	Human Validation	-0.002	0.009	0.000	-0.034	0.001
50%	Debiased	-0.002	0.009	0.000	-0.034	0.001

Table 2: Normalized Bias Summary Statistics of β_1 by Proportion of Validation Samples. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	-0.186	1.499	-0.095	-3.518	1.879
5%	Human Validation	-0.058	0.250	-0.001	-0.936	0.054
5%	Debiased	-0.068	0.425	-0.002	-1.190	0.061
10%	LLM	-0.221	1.503	-0.105	-3.684	1.869
10%	Human Validation	-0.094	0.343	-0.005	-1.344	0.048
10%	Debiased	-0.119	0.441	-0.006	-1.600	0.048
25%	LLM	-0.218	1.506	-0.111	-3.720	1.854
25%	Human Validation	-0.141	0.537	-0.001	-2.099	0.050
25%	Debiased	-0.155	0.588	-0.002	-2.256	0.056
50%	LLM	-0.215	1.505	-0.124	-3.690	1.875
50%	Human Validation	-0.201	0.769	-0.001	-3.013	0.053
50%	Debiased	-0.167	0.626	-0.003	-2.459	0.053

Table 3: Bias Summary Statistics by Model and Proportion of Validation Samples for β_1. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.000	0.014	-0.003	-0.025	0.023
GPT-3.5	5%	Human Validation	-0.002	0.008	0.000	-0.033	0.002
GPT-3.5	5%	Debiased	-0.002	0.008	0.000	-0.033	0.002
GPT-3.5	10%	LLM	-0.001	0.014	-0.003	-0.025	0.023
GPT-3.5	10%	Human Validation	-0.002	0.009	0.000	-0.034	0.001
GPT-3.5	10%	Debiased	-0.002	0.009	0.000	-0.033	0.001
GPT-3.5	25%	LLM	-0.001	0.014	-0.003	-0.025	0.023
GPT-3.5	25%	Human Validation	-0.002	0.009	0.000	-0.034	0.001
GPT-3.5	25%	Debiased	-0.002	0.009	0.000	-0.034	0.001
GPT-3.5	50%	LLM	-0.001	0.014	-0.003	-0.025	0.022
GPT-3.5	50%	Human Validation	-0.002	0.009	0.000	-0.034	0.001
GPT-3.5	50%	Debiased	-0.002	0.009	0.000	-0.034	0.001
GPT-4o	5%	LLM	-0.001	0.012	0.001	-0.034	0.016
GPT-4o	5%	Human Validation	-0.002	0.009	0.000	-0.033	0.002
GPT-4o	5%	Debiased	-0.002	0.009	0.000	-0.033	0.002
GPT-4o	10%	LLM	-0.002	0.012	0.000	-0.035	0.014
GPT-4o	10%	Human Validation	-0.002	0.009	0.000	-0.033	0.001
GPT-4o	10%	Debiased	-0.002	0.009	0.000	-0.033	0.001
GPT-4o	25%	LLM	-0.002	0.012	0.000	-0.034	0.014
GPT-4o	25%	Human Validation	-0.002	0.009	0.000	-0.034	0.001
GPT-4o	25%	Debiased	-0.002	0.009	0.000	-0.034	0.001
GPT-4o	50%	LLM	-0.002	0.012	0.000	-0.034	0.015
GPT-4o	50%	Human Validation	-0.002	0.009	0.000	-0.034	0.001
GPT-4o	50%	Debiased	-0.002	0.009	0.000	-0.034	0.001

Table 4: Normalized Bias Summary Statistics by Model and Proportion of Validation Samples for β_1. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	-0.153	1.558	-0.322	-3.297	2.213
GPT-3.5	5%	Human Validation	-0.058	0.235	-0.001	-0.935	0.054
GPT-3.5	5%	Debiased	-0.071	0.291	0.000	-1.045	0.064
GPT-3.5	10%	LLM	-0.194	1.620	-0.353	-3.343	2.179
GPT-3.5	10%	Human Validation	-0.095	0.345	-0.004	-1.356	0.042
GPT-3.5	10%	Debiased	-0.110	0.406	-0.007	-1.495	0.048
GPT-3.5	25%	LLM	-0.186	1.623	-0.330	-3.472	2.264
GPT-3.5	25%	Human Validation	-0.140	0.539	0.000	-2.099	0.049
GPT-3.5	25%	Debiased	-0.148	0.563	0.000	-2.160	0.050
GPT-3.5	50%	LLM	-0.186	1.621	-0.365	-3.357	2.172
GPT-3.5	50%	Human Validation	-0.202	0.775	0.000	-3.013	0.059
GPT-3.5	50%	Debiased	-0.165	0.620	-0.003	-2.377	0.049
GPT-4o	5%	LLM	-0.219	1.442	0.077	-4.124	1.533
GPT-4o	5%	Human Validation	-0.058	0.265	-0.002	-0.958	0.053
GPT-4o	5%	Debiased	-0.065	0.525	-0.002	-1.355	0.054
GPT-4o	10%	LLM	-0.248	1.379	0.040	-4.125	1.507
GPT-4o	10%	Human Validation	-0.092	0.341	-0.005	-1.329	0.049
GPT-4o	10%	Debiased	-0.127	0.475	-0.003	-1.851	0.048
GPT-4o	25%	LLM	-0.250	1.384	0.049	-4.148	1.463
GPT-4o	25%	Human Validation	-0.142	0.536	-0.004	-2.077	0.051
GPT-4o	25%	Debiased	-0.162	0.613	-0.004	-2.364	0.060
GPT-4o	50%	LLM	-0.243	1.384	0.057	-4.166	1.510
GPT-4o	50%	Human Validation	-0.200	0.766	-0.002	-2.989	0.051
GPT-4o	50%	Debiased	-0.169	0.633	-0.005	-2.490	0.055

Table 5: MSE Summary Statistics of β_1 by Proportion of Validation Samples. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.000	0.001	0.000	0.000	0.002
5%	Human Validation	0.002	0.001	0.002	0.001	0.004
5%	Debiased	0.002	0.001	0.001	0.001	0.003
10%	LLM	0.000	0.001	0.000	0.000	0.002
10%	Human Validation	0.001	0.001	0.001	0.000	0.003
10%	Debiased	0.001	0.001	0.001	0.000	0.003
25%	LLM	0.000	0.001	0.000	0.000	0.002
25%	Human Validation	0.001	0.001	0.000	0.000	0.003
25%	Debiased	0.001	0.001	0.000	0.000	0.003
50%	LLM	0.000	0.001	0.000	0.000	0.002
50%	Human Validation	0.000	0.001	0.000	0.000	0.002
50%	Debiased	0.000	0.001	0.000	0.000	0.002

Table 6: MSE Summary Statistics by Model and Proportion of Validation Samples for β_1. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.000	0.000	0.000	0.000	0.002
GPT-3.5	5%	Human Validation	0.002	0.001	0.002	0.001	0.004
GPT-3.5	5%	Debiased	0.002	0.001	0.002	0.001	0.003
GPT-3.5	10%	LLM	0.000	0.001	0.000	0.000	0.002
GPT-3.5	10%	Human Validation	0.001	0.001	0.001	0.000	0.003
GPT-3.5	10%	Debiased	0.001	0.001	0.001	0.000	0.003
GPT-3.5	25%	LLM	0.000	0.001	0.000	0.000	0.002
GPT-3.5	25%	Human Validation	0.001	0.001	0.000	0.000	0.003
GPT-3.5	25%	Debiased	0.001	0.001	0.000	0.000	0.003
GPT-3.5	50%	LLM	0.000	0.001	0.000	0.000	0.002
GPT-3.5	50%	Human Validation	0.000	0.001	0.000	0.000	0.002
GPT-3.5	50%	Debiased	0.000	0.001	0.000	0.000	0.002
GPT-4o	5%	LLM	0.000	0.001	0.000	0.000	0.002
GPT-4o	5%	Human Validation	0.002	0.001	0.002	0.001	0.004
GPT-4o	5%	Debiased	0.001	0.001	0.001	0.001	0.003
GPT-4o	10%	LLM	0.000	0.001	0.000	0.000	0.002
GPT-4o	10%	Human Validation	0.001	0.001	0.001	0.000	0.003
GPT-4o	10%	Debiased	0.001	0.001	0.001	0.000	0.003
GPT-4o	25%	LLM	0.000	0.001	0.000	0.000	0.002
GPT-4o	25%	Human Validation	0.001	0.001	0.000	0.000	0.003
GPT-4o	25%	Debiased	0.001	0.001	0.000	0.000	0.003
GPT-4o	50%	LLM	0.000	0.001	0.000	0.000	0.002
GPT-4o	50%	Human Validation	0.000	0.001	0.000	0.000	0.002
GPT-4o	50%	Debiased	0.000	0.001	0.000	0.000	0.002

Table 7: Coverage Summary Statistics of β_1 by Proportion of Validation Samples. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

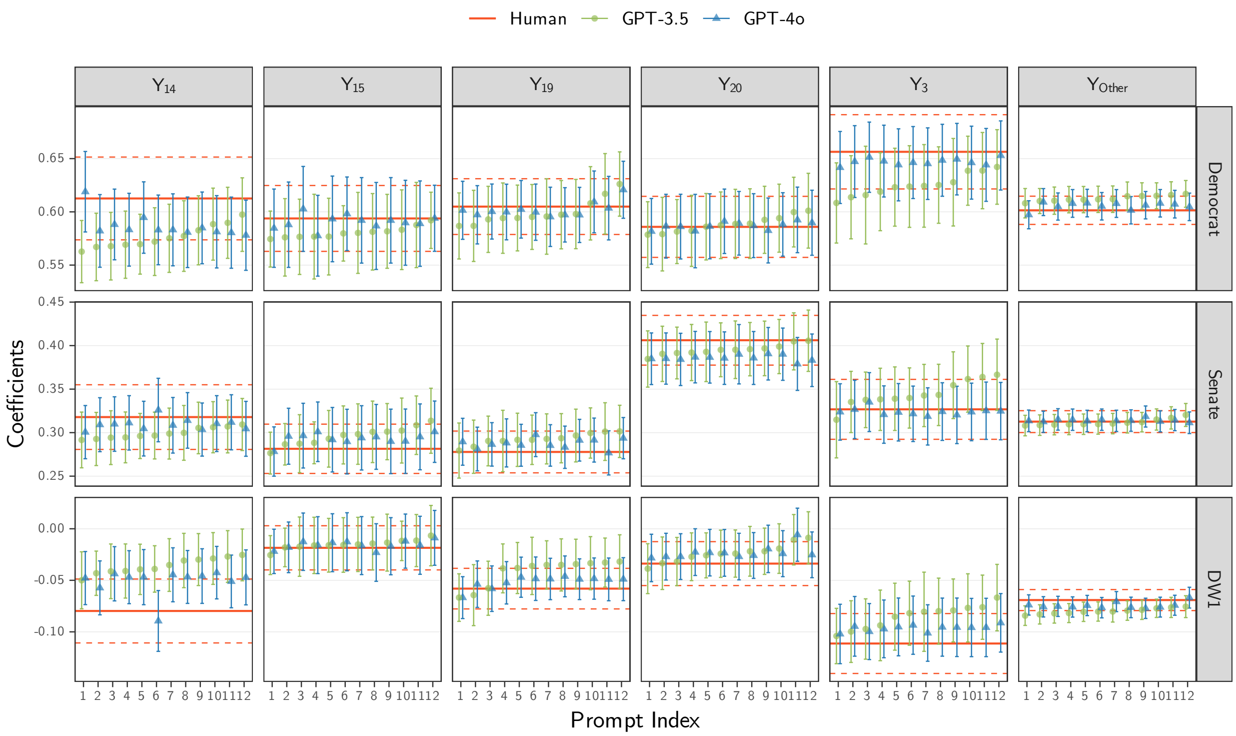
Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.780	0.183	0.856	0.416	0.950
5%	Human Validation	0.930	0.060	0.946	0.721	0.959
5%	Debiased	0.906	0.088	0.929	0.628	0.945
10%	LLM	0.779	0.184	0.855	0.422	0.951
10%	Human Validation	0.924	0.089	0.948	0.597	0.959
10%	Debiased	0.912	0.108	0.941	0.525	0.952
25%	LLM	0.779	0.183	0.854	0.406	0.952
25%	Human Validation	0.918	0.117	0.948	0.484	0.959
25%	Debiased	0.915	0.118	0.946	0.478	0.957
50%	LLM	0.780	0.183	0.855	0.418	0.950
50%	Human Validation	0.918	0.119	0.949	0.478	0.961
50%	Debiased	0.916	0.118	0.948	0.476	0.959

Table 8: Coverage Summary Statistics by Model and Proportion of Validation Samples for β_1. Proxy on the LHS: $Y = \beta_0 + \beta_1 V$.

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.733	0.200	0.780	0.375	0.945
GPT-3.5	5%	Human Validation	0.933	0.056	0.947	0.727	0.961
GPT-3.5	5%	Debiased	0.911	0.075	0.930	0.662	0.945
GPT-3.5	10%	LLM	0.730	0.202	0.792	0.376	0.949
GPT-3.5	10%	Human Validation	0.923	0.089	0.947	0.597	0.959
GPT-3.5	10%	Debiased	0.913	0.104	0.941	0.545	0.952
GPT-3.5	25%	LLM	0.731	0.201	0.780	0.362	0.944
GPT-3.5	25%	Human Validation	0.917	0.117	0.948	0.483	0.958
GPT-3.5	25%	Debiased	0.915	0.118	0.946	0.478	0.957
GPT-3.5	50%	LLM	0.731	0.202	0.786	0.369	0.947
GPT-3.5	50%	Human Validation	0.919	0.119	0.950	0.479	0.961
GPT-3.5	50%	Debiased	0.916	0.119	0.948	0.474	0.959
GPT-4o	5%	LLM	0.827	0.150	0.905	0.474	0.951
GPT-4o	5%	Human Validation	0.928	0.063	0.945	0.717	0.957
GPT-4o	5%	Debiased	0.901	0.099	0.927	0.560	0.945
GPT-4o	10%	LLM	0.828	0.150	0.905	0.471	0.952
GPT-4o	10%	Human Validation	0.925	0.089	0.949	0.597	0.959
GPT-4o	10%	Debiased	0.910	0.112	0.940	0.496	0.953
GPT-4o	25%	LLM	0.827	0.149	0.904	0.475	0.954
GPT-4o	25%	Human Validation	0.918	0.117	0.948	0.484	0.960
GPT-4o	25%	Debiased	0.914	0.119	0.945	0.475	0.957
GPT-4o	50%	LLM	0.828	0.147	0.904	0.475	0.950
GPT-4o	50%	Human Validation	0.918	0.119	0.949	0.478	0.961
GPT-4o	50%	Debiased	0.916	0.118	0.948	0.478	0.959

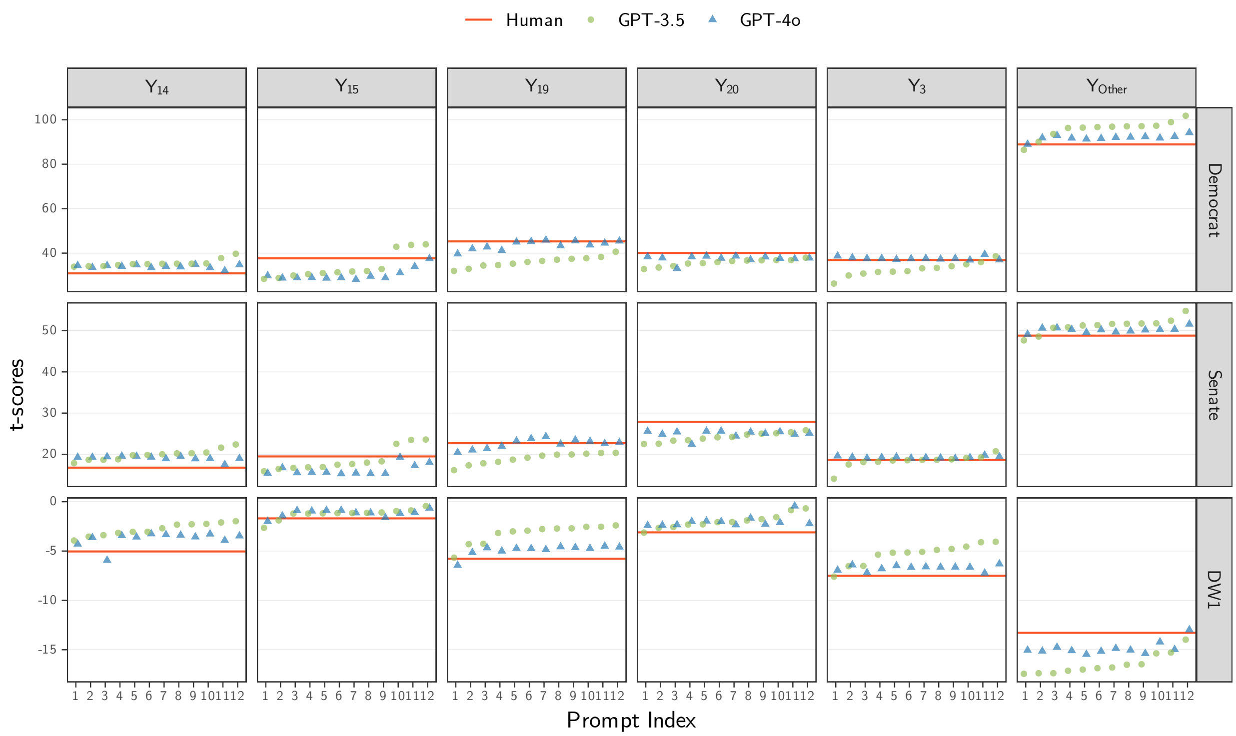
Proxy on the RHS

Figure 10: β_t vs. Prompt for all βs. Proxy on the RHS.



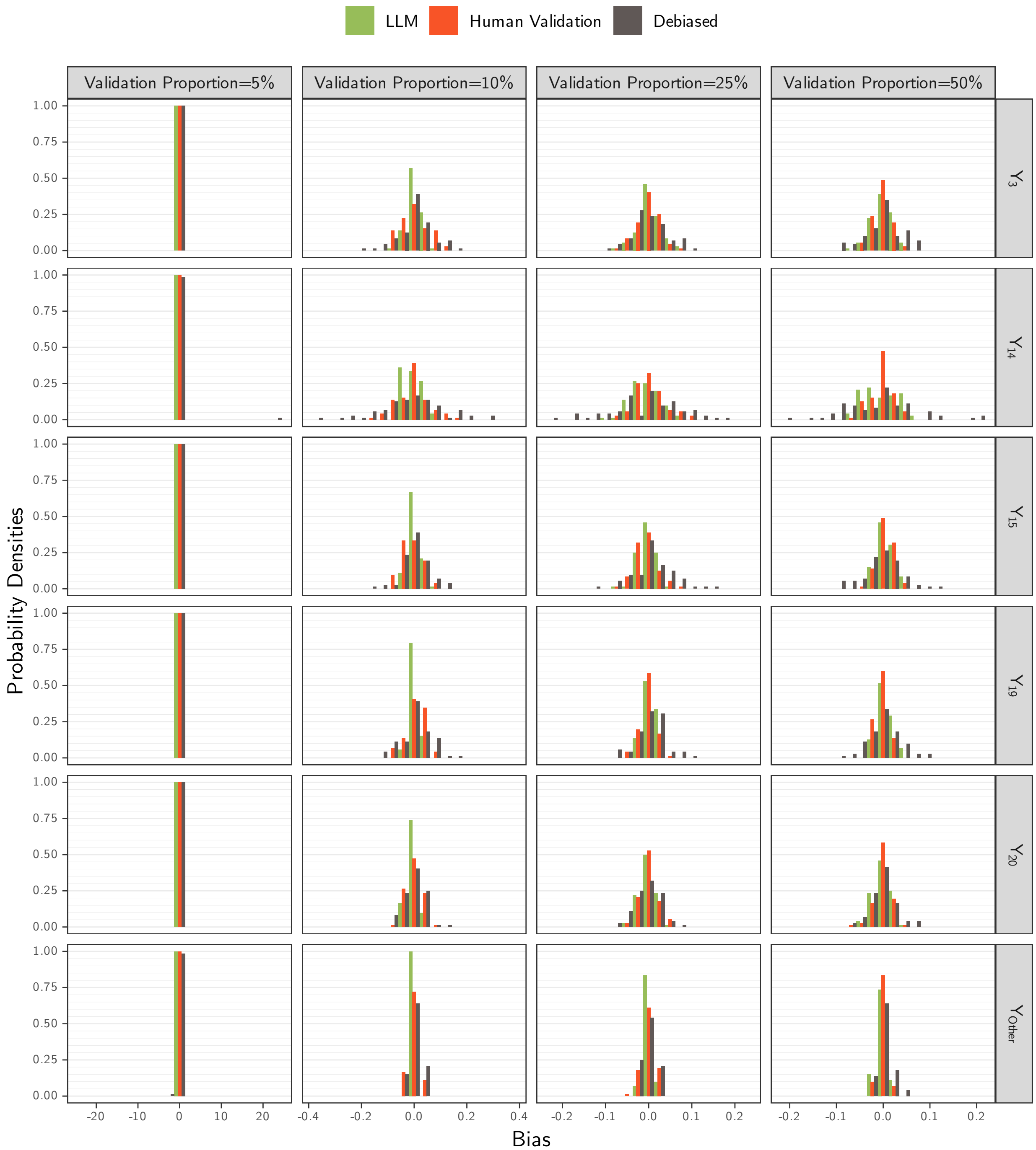
Note: The coefficients were obtained by fitting the model $V = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$ over 10,000 bills, with 95% confidence intervals computed using robust standard errors. Here, Y_t denotes the indicator for topic t, where $Y_t = 1\{Y = t\}$ for $t \in \mathcal{T}$ and $Y_{\text{Other}} = 1\{Y \notin \mathcal{T}\}$, with $\mathcal{T} = \{3, 14, 15, 19, 20\}$ is the 5 most-common human-labeled topics in the bills data. The variable V takes one of the following bills variables: Democrat, Senate, or DW1, where $extDemocrat = 1\{\text{Party} = \text{Democrat}\}$, Senate $= 1\{\text{Chamber} = \text{Senate}\}$, and DW1 is the imputed DW1 score for the bill sponsor. The regressions based on human-labeled topics (red) were estimated with $Y_t^{\text{Human}} = 1\{Y^{\text{Human}} = t\}$, while the LLM-based regressions (GPT-3.5 in green, circles; and GPT-4o in blue, triangles) were estimated with $Y_{t,m,p}^{\text{LLM}} = 1\{Y_{m,p}^{\text{LLM}} = t\}$, where $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$ denotes the LLM model and $p \in \{1, \ldots, 12\}$ denotes the prompt modifications. In each subplot, the indices for the prompts reflect the order chosen based on sorting the estimated values of the GPT-3.5 model in ascending order. The corresponding estimates for GPT-4o are aligned according to this order.

Figure 11: t-scores of each β_t vs. Prompt. Proxy on the RHS.



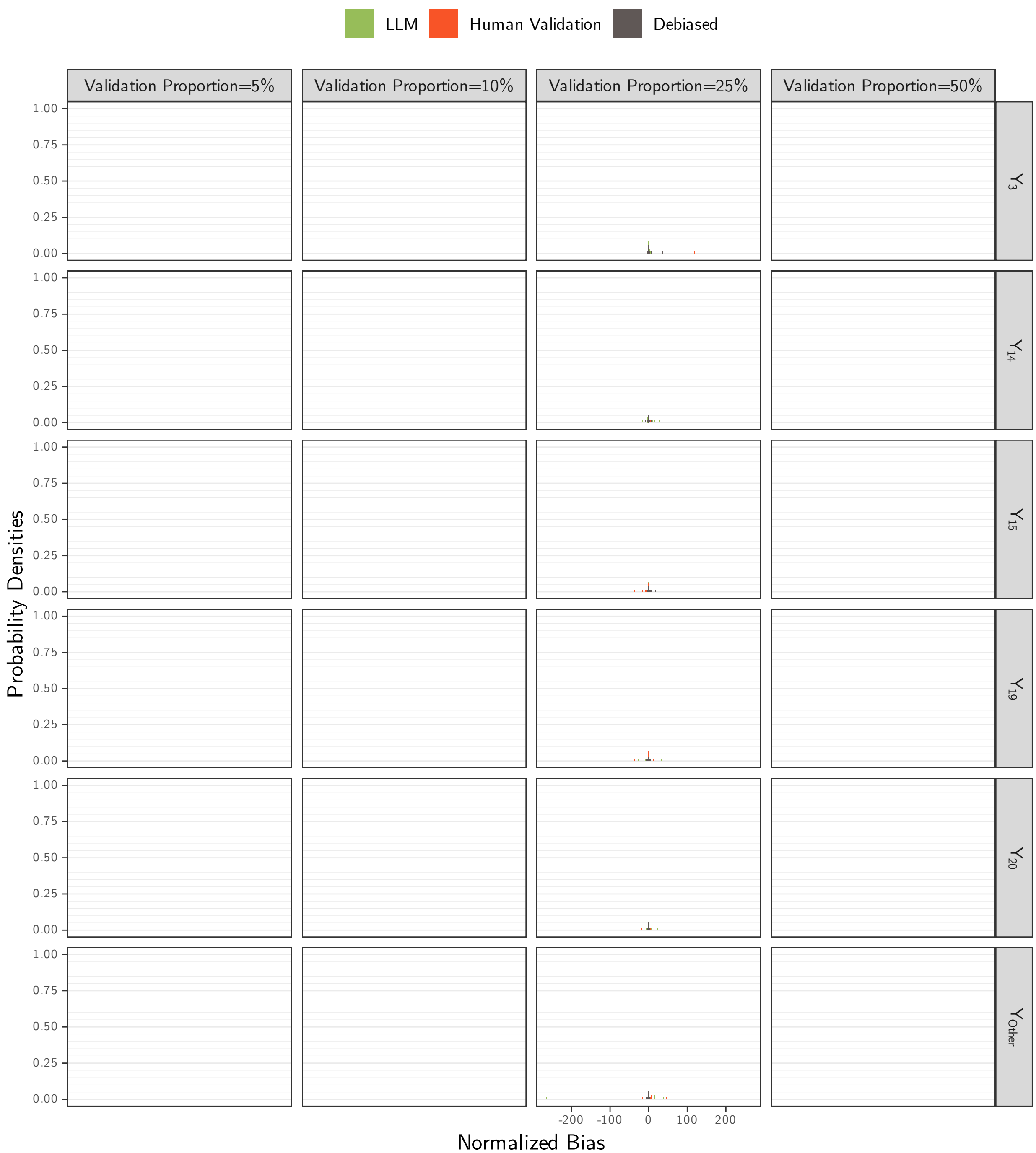
Note: The t-scores were obtained by fitting the model $V = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$ over 10,000 bills, with 95% confidence intervals computed using robust standard errors. Here, Y_t denotes the indicator for topic t, where $Y_t = 1\{Y = t\}$ for $t \in \mathcal{T}$ and $Y_{\text{Other}} = 1\{Y \notin \mathcal{T}\}$, with $\mathcal{T} = \{3, 14, 15, 19, 20\}$ is the 5 most-common human-labeled topics in the bills data. The variable V takes one of the following bills variables: Democrat, Senate, or DW1, where $extDemocrat = 1\{\text{Party} = \text{Democrat}\}$, Senate $= 1\{\text{Chamber} = \text{Senate}\}$, and DW1 is the imputed DW1 score for the bill sponsor. The regressions based on human-labeled topics (red) were estimated with $Y_t^{\text{Human}} = 1\{Y^{\text{Human}} = t\}$, while the LLM-based regressions (GPT-3.5 in green, circles; and GPT-4o in blue, triangles) were estimated with $Y_{t,m,p}^{\text{LLM}} = 1\{Y_{m,p}^{\text{LLM}} = t\}$, where $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$ denotes the LLM model and $p \in \{1, \ldots, 12\}$ denotes the prompt modifications. In each subplot, the indices for the prompts reflect the order chosen based on sorting the estimated values of the GPT-3.5 model in ascending order. The corresponding estimates for GPT-4o are aligned according to this order.

Figure 12: Bias Density Plots for all βs by Proportion of Validation Samples. Proxy on the RHS.



Note: The bias estimates were computed as the difference between the coefficient estimates from $N = 1000$ simulation runs, $\bar{\beta}_{V,\eta,m,p} = \sum_{i=1}^{N} \beta^{(i)}_{t,V,\eta,m,p}/N$, and the human-based coefficient from the entire set of 10,000 bills, $\beta^{\star}_{t,V}$, using a regression model of the form $V = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$, for topic $t \in \{3, 14, 15, 19, 20, \text{Other}\}$, variable $V \in \{\text{Democrat}, \text{Senate}, \text{DW1}\}$, validation proportion $\eta \in \{5\%, 10\%, 25\%, 50\%\}$, LLM model $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$, and prompt $p \in \{1, \ldots, 12\}$. For the LLM-based models (green), a random sample of 5,000 bills drawn from the original 10,000 was used, with LLM-based labels, $Y^{\text{LLM}}_{t,m,p} = 1\{Y^{\text{LLM}}_{m,p} = t\}$. For the other two models—human-based (red) and debiased-based (brown)—the 5,000 sample were split into two subsets: validation and primary, with the size of the validation set determined by the validation proportion, η. In the human-based model, only the human labels in the validation set were used, $Y^{\text{Human}}_t = 1\{Y^{\text{Human}} = t\}$, whereas the debiased estimates use both the validation and primary sets, with Y^{Human}_t from validation set used to estimate measurement errors and $Y^{\text{LLM}}_{t,m,p}$ from the primary set used to obtain the coefficient estimates used to compute the reported bias.

Figure 13: Normalized Bias Density of β_1 by Proportion of Validation Samples. Proxy on the LHS.



Note: The normalized bias estimates were computed as the normalized difference between the coefficient estimates from $N = 1000$ simulation runs, $\bar{\beta}_{t,V,\eta,m,p}$, and the human-based coefficient from the entire set of 10,000 bills, $\beta^{\star}_{t,V}$, divided by the sample standard deviation, $s_{t,V,\eta,m,p}$, where $s^2_{t,V,\eta,m,p} = \sum_{i=1}^{N}(\beta^{(i)}_{t,V,\eta,m,p} - \bar{\beta}_{t,V,\eta,m,p})^2/(N-1)$ and all variables are as defined previously.

Figure 14: Bias Density of β_3 by Model and Proportion of Validation Samples. Proxy on the RHS.

Figure 15: Bias Density of β_{14} by Model and Proportion of Validation Samples. Proxy on the RHS.

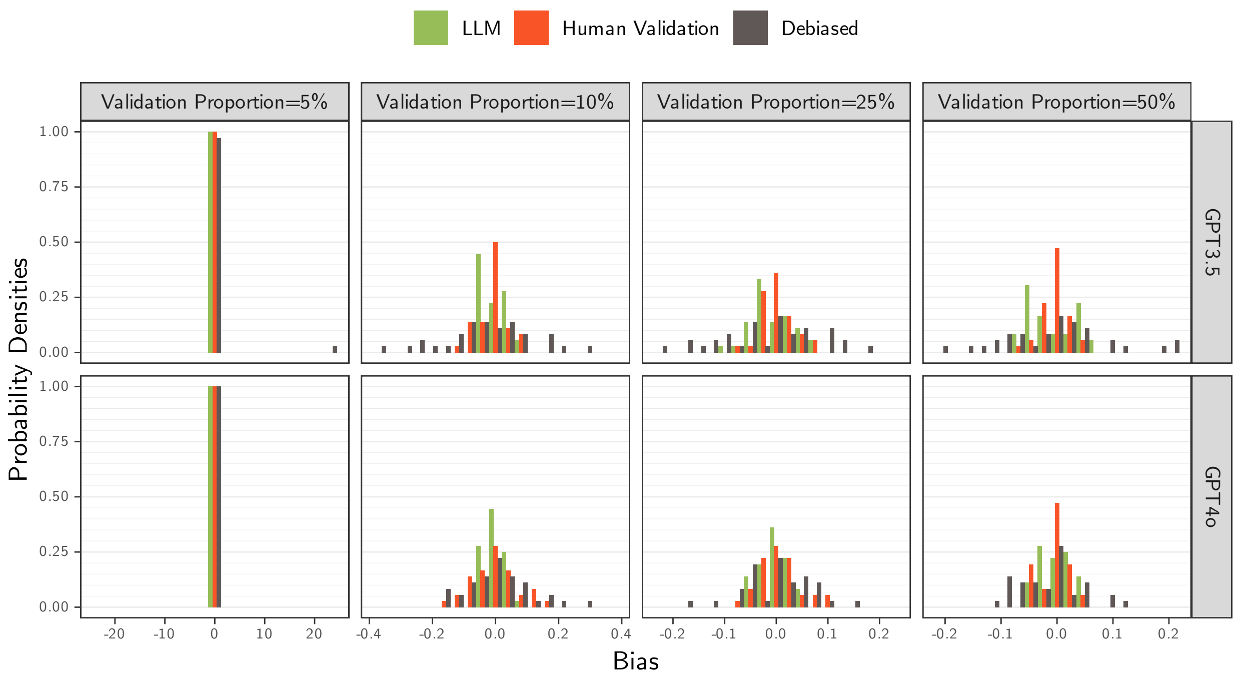


Figure 16: Bias Density of β_{15} by Model and Proportion of Validation Samples. Proxy on the RHS.

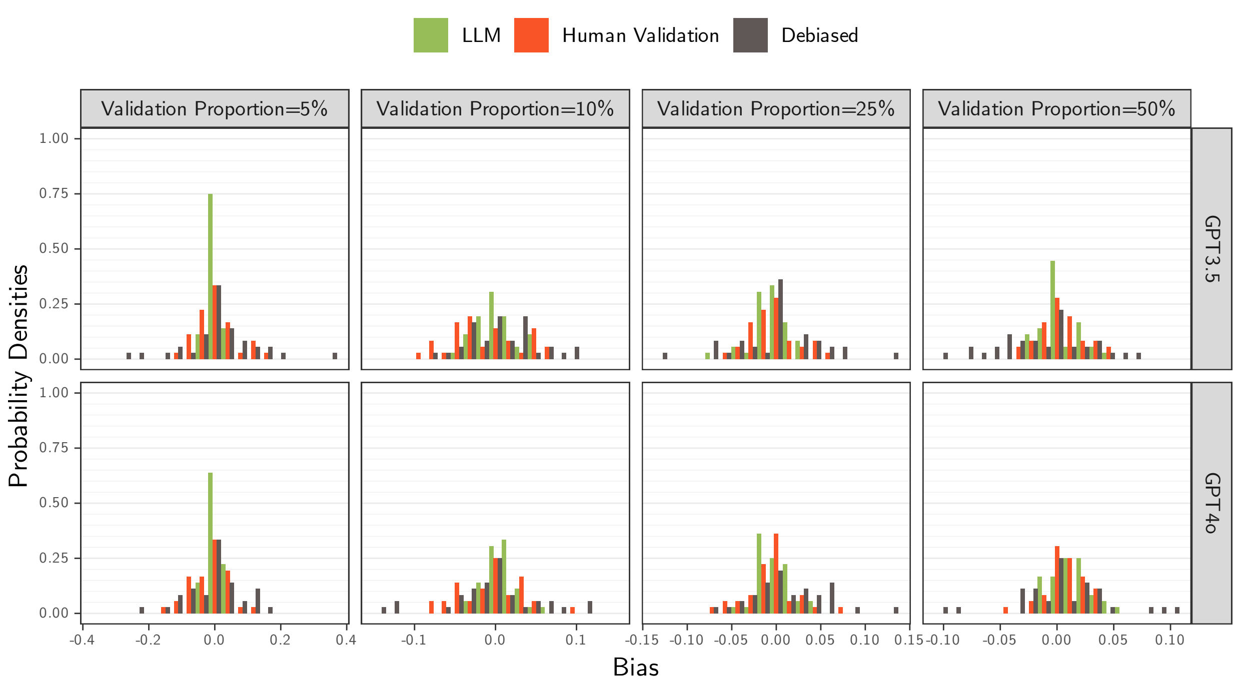


Figure 17: Bias Density of β_{19} by Model and Proportion of Validation Samples. Proxy on the RHS.

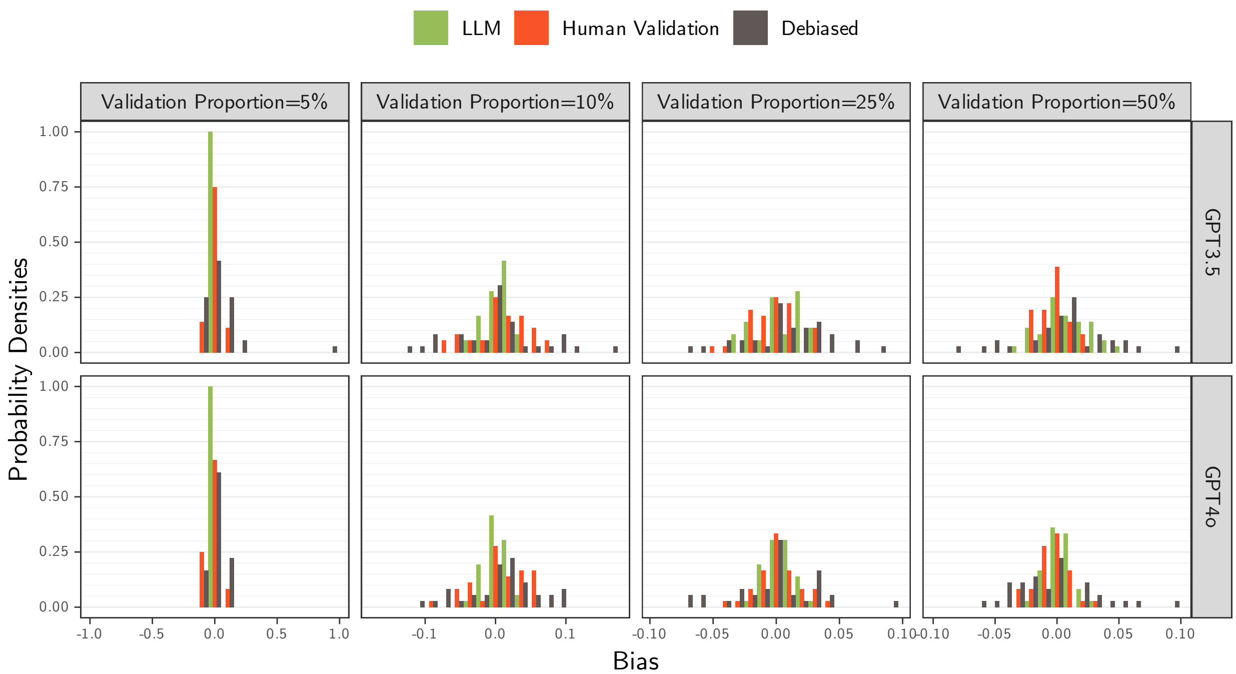


Figure 18: Bias Density of β_{20} by Model and Proportion of Validation Samples. Proxy on the RHS.

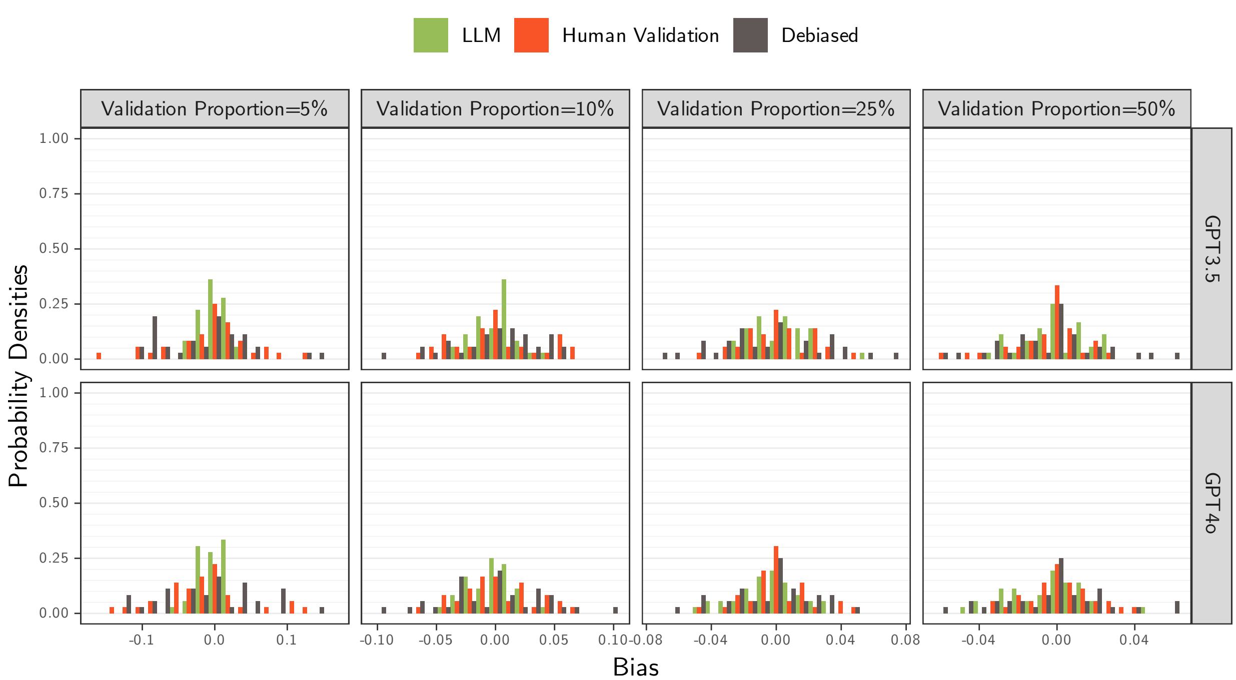


Figure 19: Bias Density of β_{Other} by Model and Proportion of Validation Samples. Proxy on the RHS.

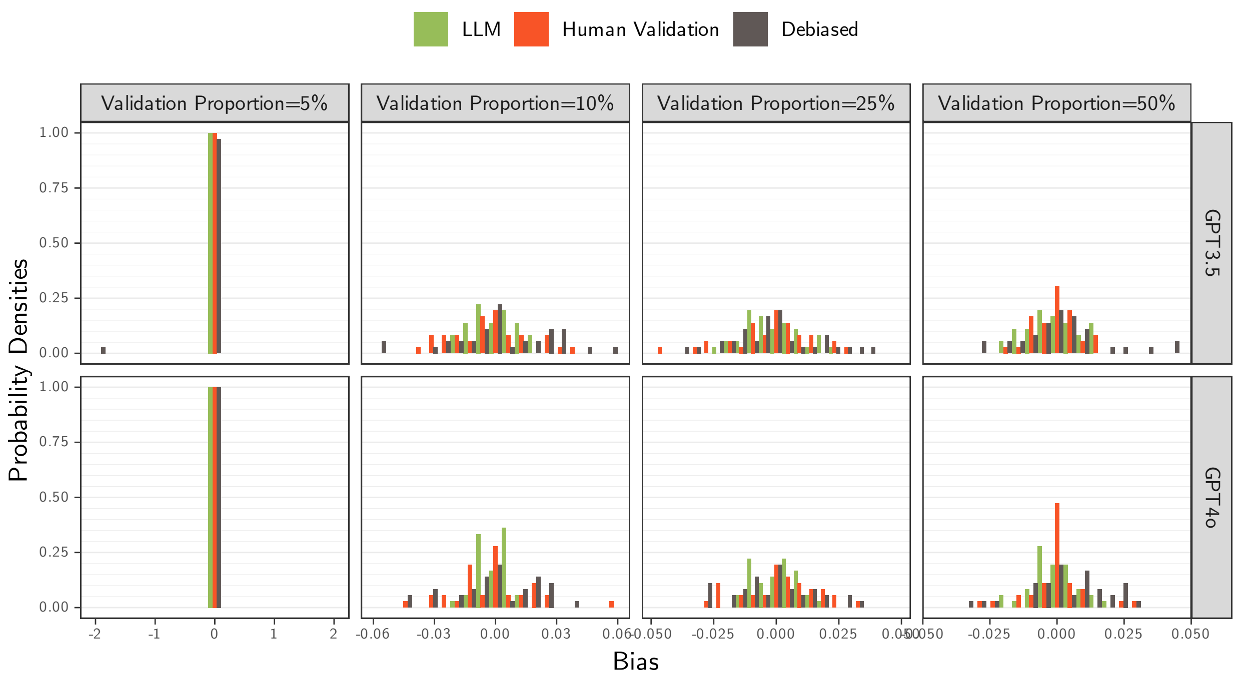


Figure 20: Normalized Bias Density of β_3 by Model and Proportion of Validation Samples. Proxy on the RHS.

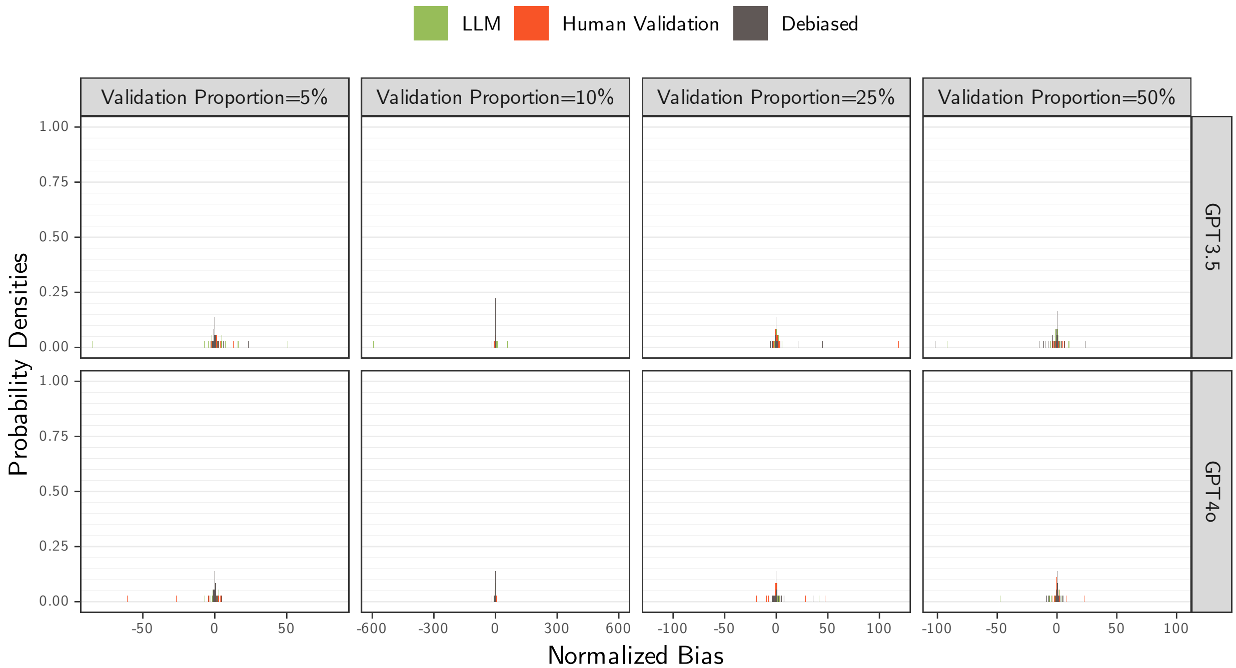


Figure 21: Normalized Bias Density of β_{14} by Model and Proportion of Validation Samples. Proxy on the RHS.

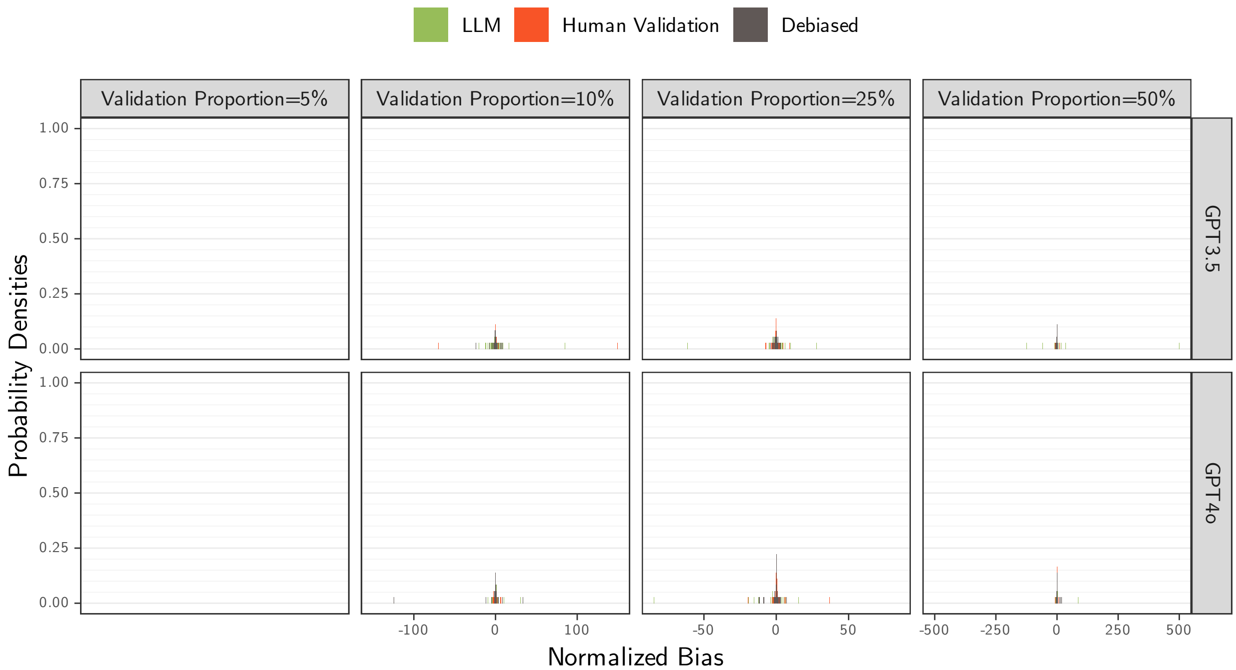


Figure 22: Normalized Bias Density of β_{15} by Model and Proportion of Validation Samples. Proxy on the RHS.

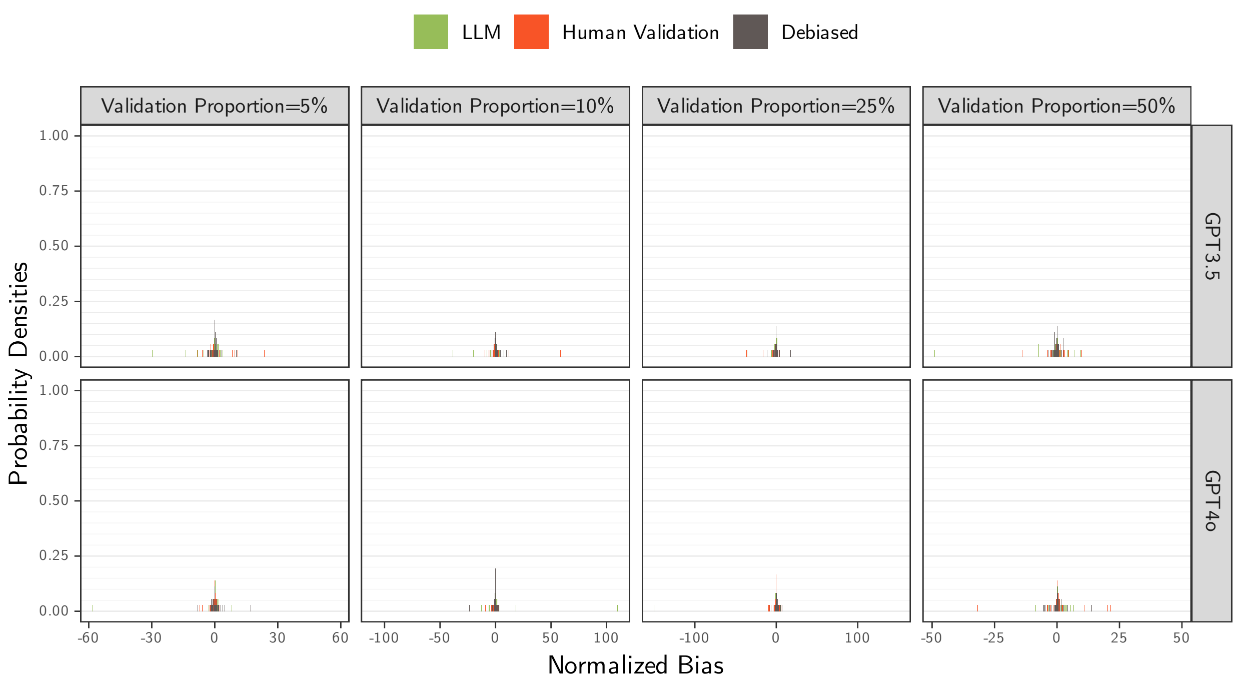


Figure 23: Normalized Bias Density of β_{19} by Model and Proportion of Validation Samples. Proxy on the RHS.

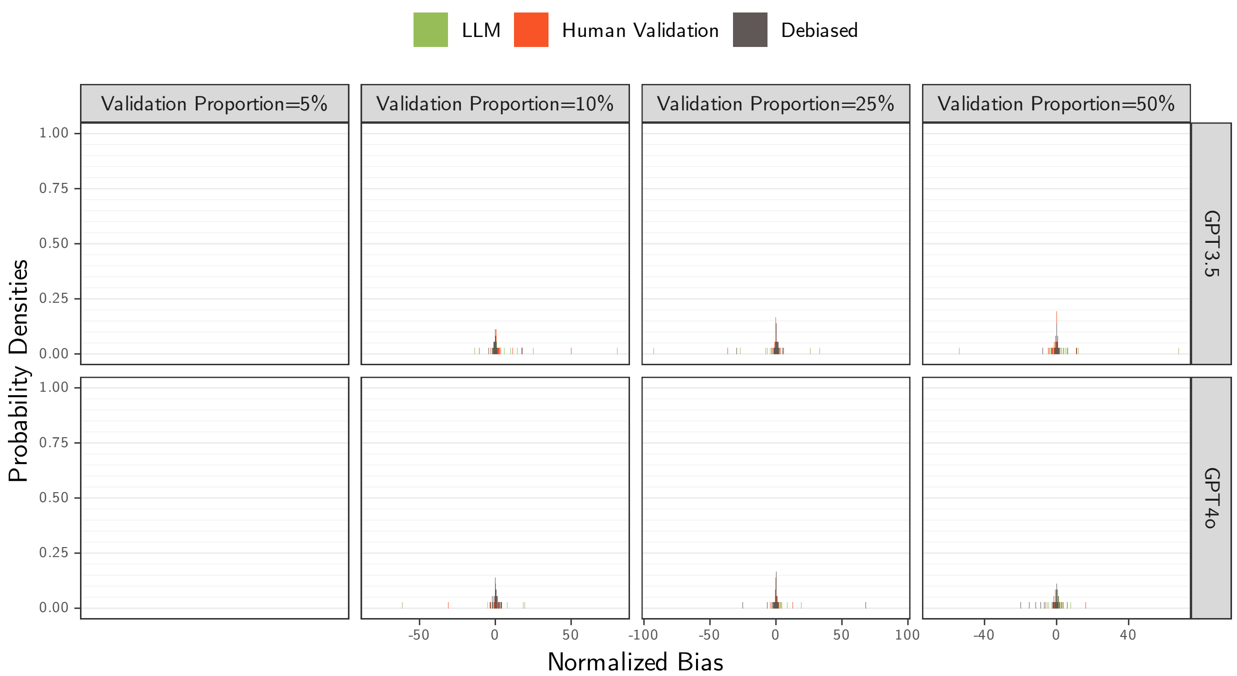


Figure 24: Normalized Bias Density of β_{20} by Model and Proportion of Validation Samples. Proxy on the RHS.

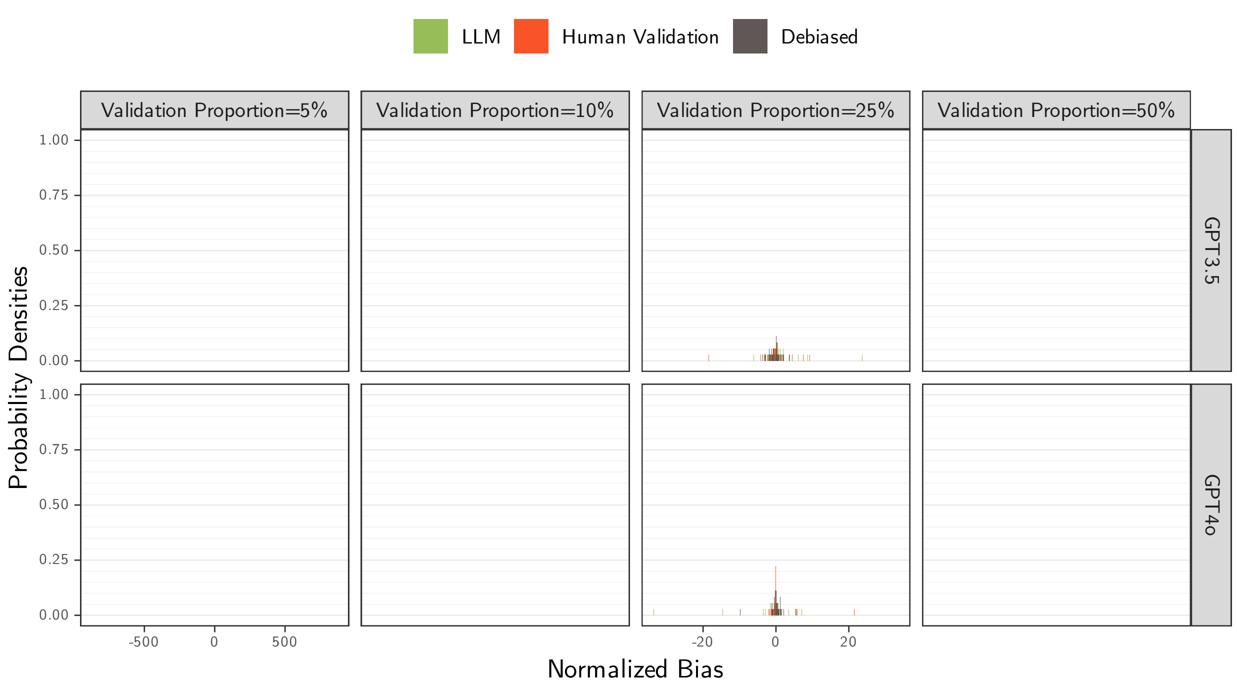


Figure 25: Normalized Bias Density of β_{Other} by Model and Proportion of Validation Samples. Proxy on the RHS.

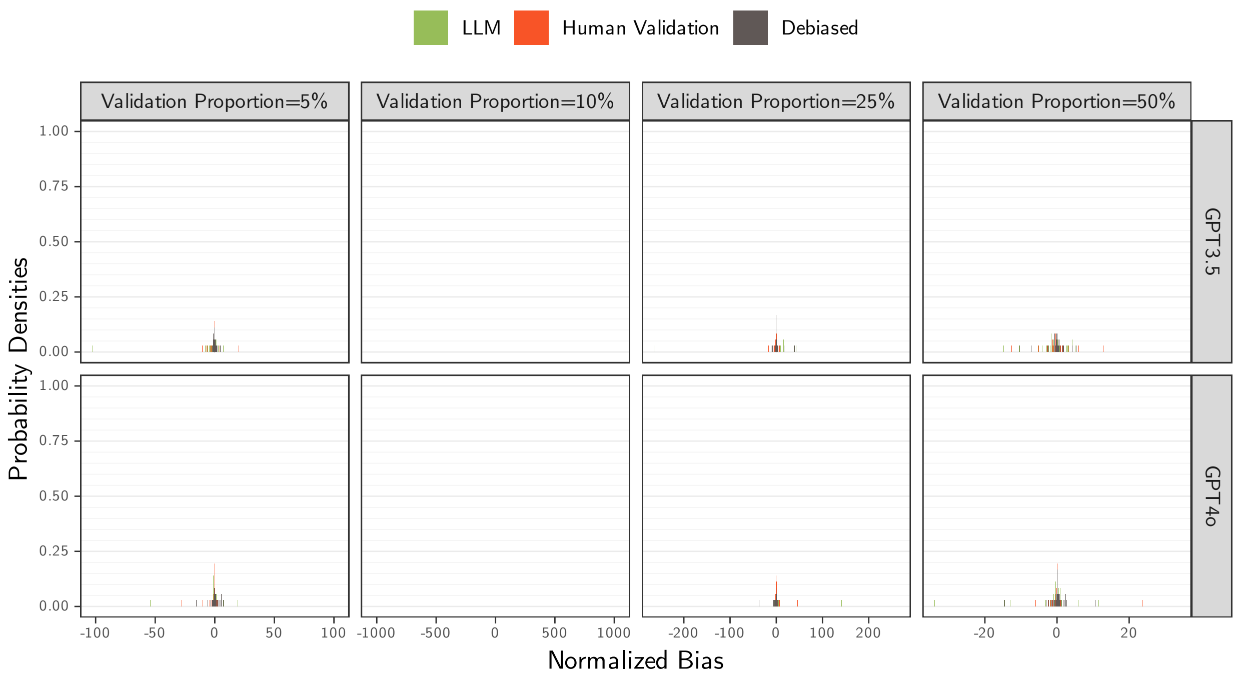


Figure 26: MSE Empirical CDF for all βs by Proportion of Validation Samples. Proxy on the RHS.

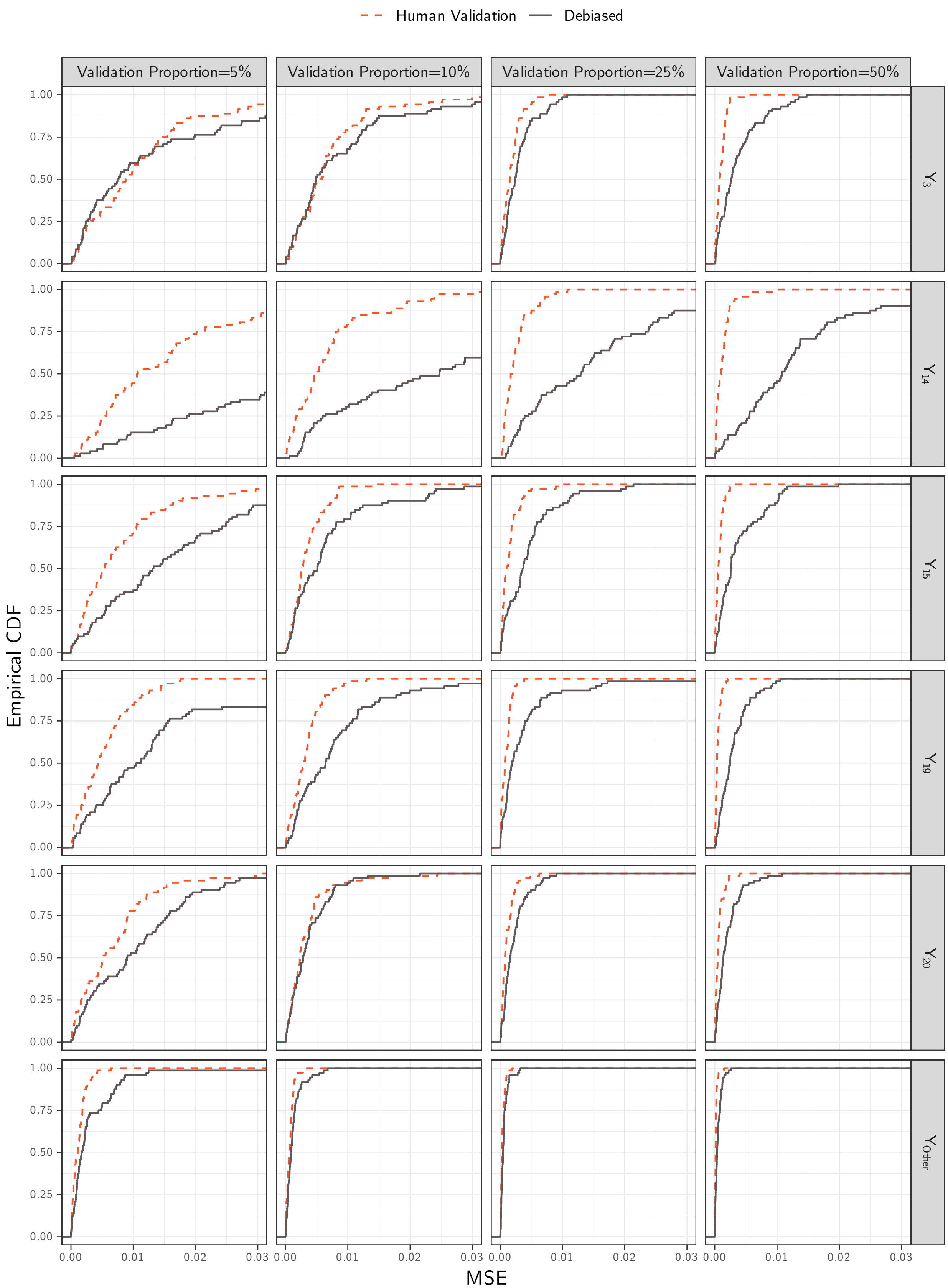


Figure 27: MSE Empirical CDF of β_3 by Model and Proportion of Validation Samples. Proxy on the RHS.

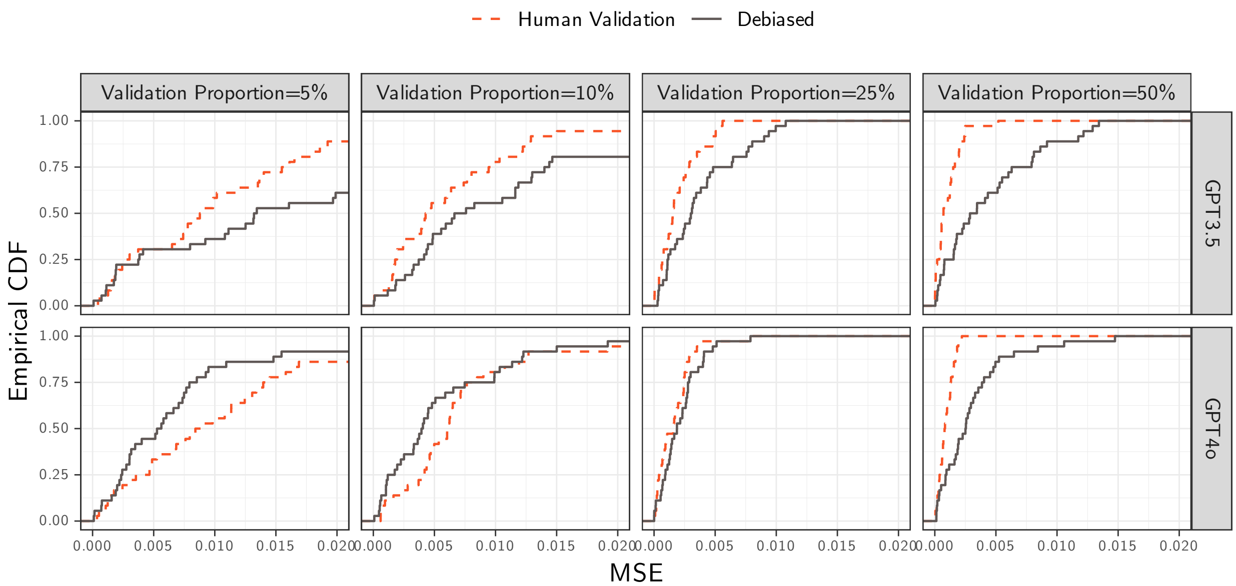


Figure 28: MSE Empirical CDF of β_{14} by Model and Proportion of Validation Samples. Proxy on the RHS.

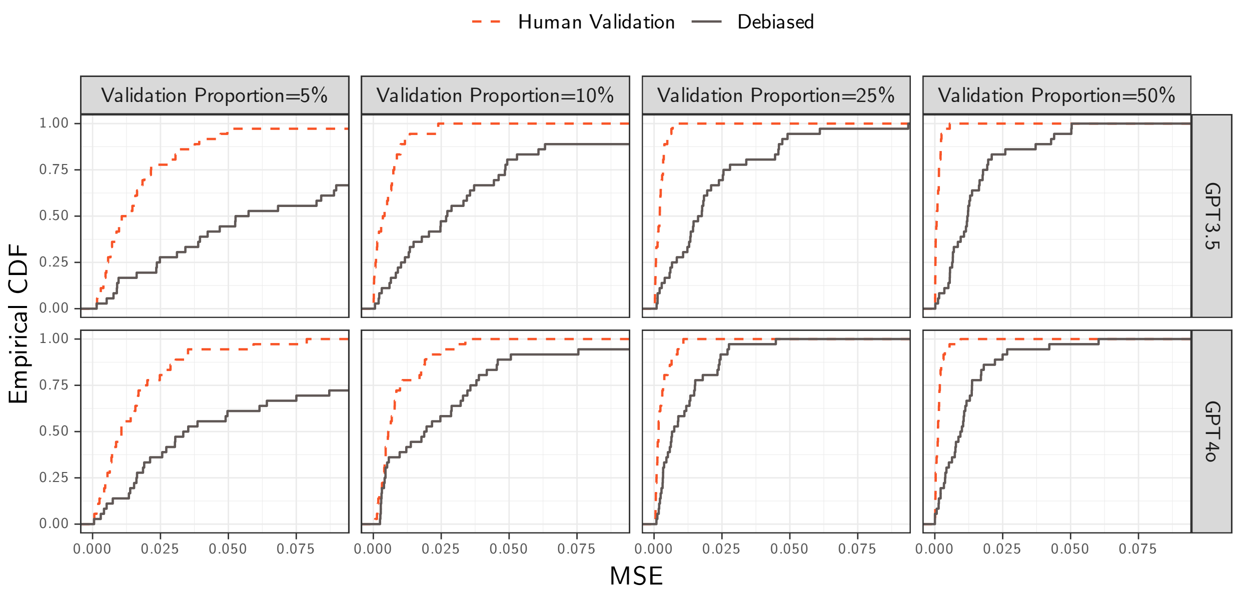


Figure 29: MSE Empirical CDF of β_{15} by Model and Proportion of Validation Samples. Proxy on the RHS.

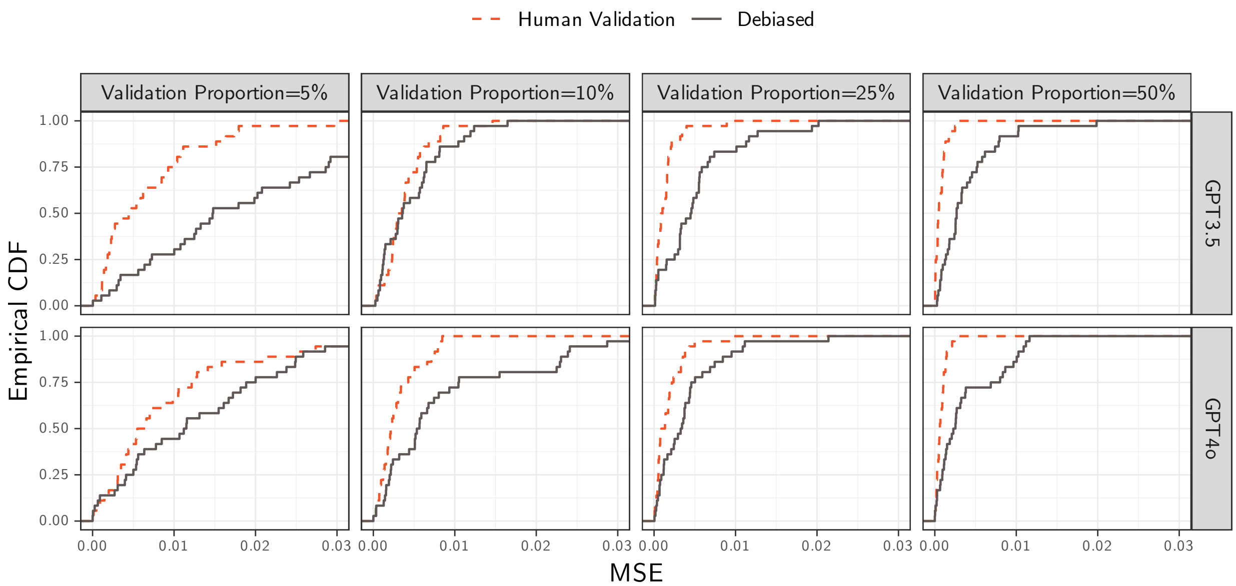


Figure 30: MSE Empirical CDF of β_{19} by Model and Proportion of Validation Samples. Proxy on the RHS.

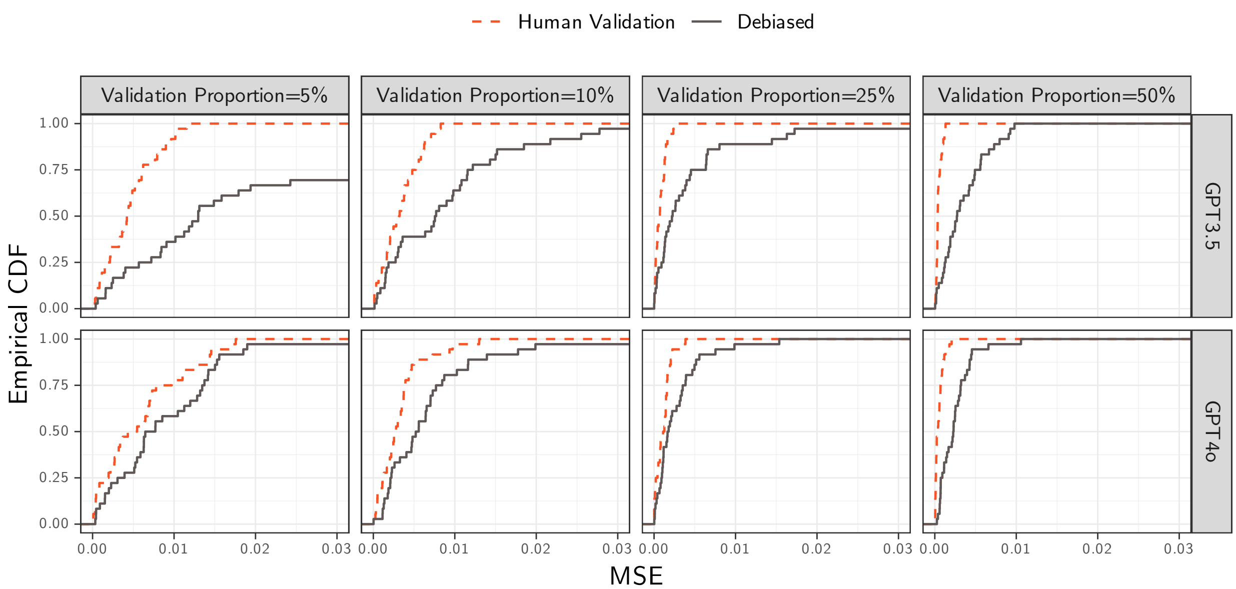


Figure 31: MSE Empirical CDF of β_{20} by Model and Proportion of Validation Samples. Proxy on the RHS.

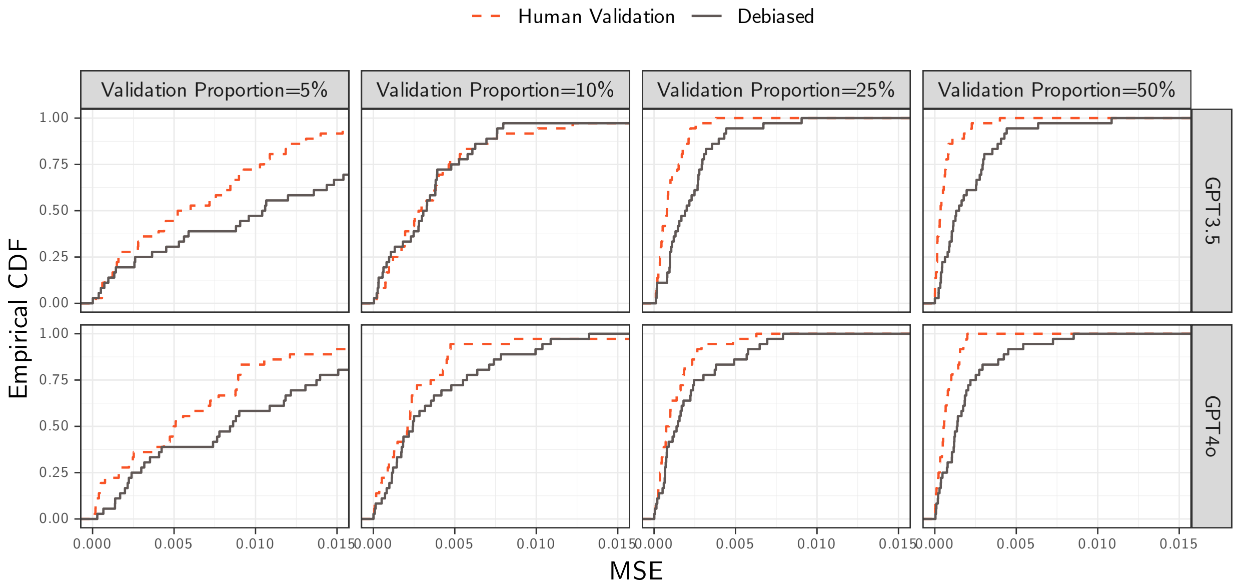


Figure 32: MSE Empirical CDF of β_{Other} by Model and Proportion of Validation Samples. Proxy on the RHS.

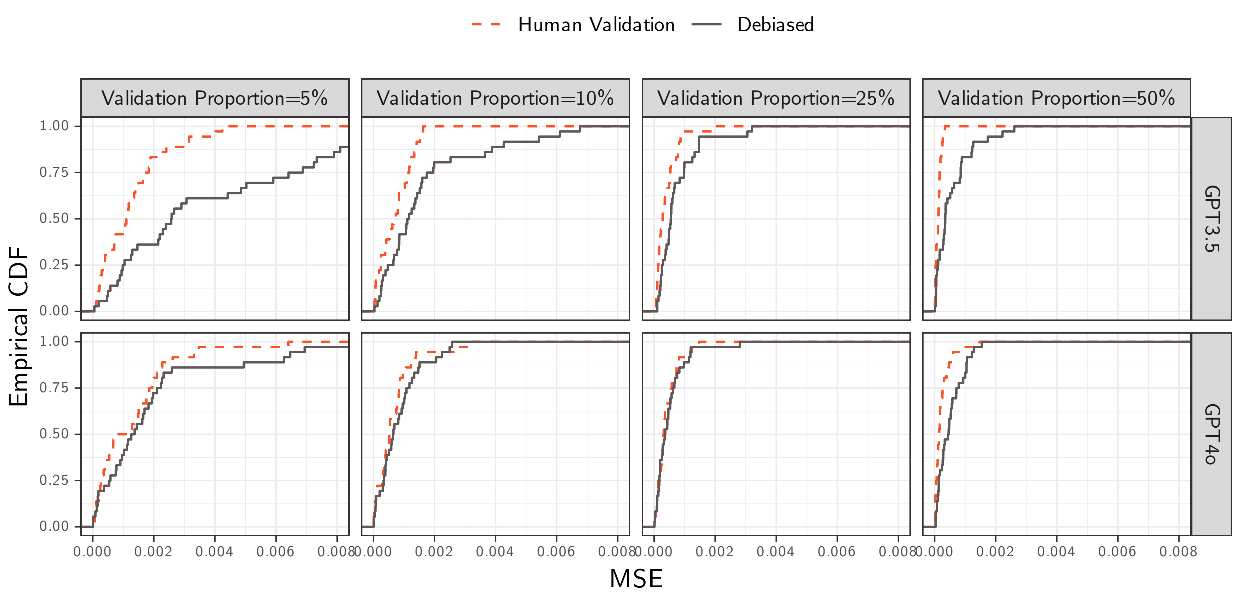


Table 9: Bias Summary Statistics of β_3 by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	0.002	0.026	0.003	-0.044	0.040
5%	Human Validation	0.012	0.075	0.014	-0.089	0.121
5%	Debiased	-0.007	0.084	0.000	-0.164	0.111
10%	LLM	0.005	0.026	0.006	-0.040	0.040
10%	Human Validation	0.001	0.053	-0.003	-0.073	0.099
10%	Debiased	0.001	0.066	0.000	-0.105	0.131
25%	LLM	0.004	0.027	0.004	-0.046	0.052
25%	Human Validation	-0.001	0.028	0.000	-0.042	0.038
25%	Debiased	0.000	0.040	-0.001	-0.061	0.070
50%	LLM	0.001	0.024	0.003	-0.045	0.036
50%	Human Validation	-0.002	0.019	-0.002	-0.035	0.030
50%	Debiased	-0.003	0.039	0.000	-0.079	0.058

Table 10: Bias Summary Statistics of β_{14} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	-0.001	0.035	-0.007	-0.051	0.061
5%	Human Validation	-0.014	0.087	-0.008	-0.150	0.136
5%	Debiased	0.371	2.889	0.031	-0.251	0.409
10%	LLM	-0.001	0.034	-0.010	-0.043	0.056
10%	Human Validation	-0.005	0.060	-0.002	-0.100	0.092
10%	Debiased	-0.014	0.125	-0.005	-0.223	0.194
25%	LLM	-0.004	0.035	-0.008	-0.055	0.050
25%	Human Validation	0.003	0.036	0.002	-0.049	0.068
25%	Debiased	-0.007	0.081	-0.001	-0.156	0.107
50%	LLM	-0.004	0.036	-0.009	-0.054	0.050
50%	Human Validation	-0.004	0.024	0.001	-0.046	0.031
50%	Debiased	-0.009	0.079	-0.004	-0.110	0.113

Table 11: Bias Summary Statistics of β_{15} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	0.002	0.018	0.002	-0.025	0.032
5%	Human Validation	-0.007	0.060	-0.003	-0.097	0.104
5%	Debiased	0.002	0.100	0.002	-0.146	0.148
10%	LLM	0.005	0.022	0.005	-0.028	0.047
10%	Human Validation	-0.010	0.041	-0.004	-0.075	0.054
10%	Debiased	0.001	0.051	-0.001	-0.070	0.091
25%	LLM	-0.002	0.020	-0.002	-0.033	0.031
25%	Human Validation	-0.006	0.026	-0.005	-0.052	0.038
25%	Debiased	0.006	0.047	0.001	-0.066	0.076
50%	LLM	0.009	0.017	0.005	-0.017	0.041
50%	Human Validation	0.005	0.018	0.004	-0.021	0.032
50%	Debiased	-0.005	0.041	-0.002	-0.078	0.061

Table 12: Bias Summary Statistics of β_{19} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	0.005	0.019	0.005	-0.024	0.033
5%	Human Validation	-0.009	0.050	-0.001	-0.092	0.088
5%	Debiased	0.021	0.135	-0.002	-0.097	0.145
10%	LLM	0.005	0.016	0.007	-0.021	0.028
10%	Human Validation	0.007	0.039	0.006	-0.062	0.060
10%	Debiased	0.004	0.058	0.003	-0.096	0.098
25%	LLM	0.004	0.016	0.004	-0.024	0.027
25%	Human Validation	0.000	0.017	0.000	-0.027	0.031
25%	Debiased	0.003	0.034	0.001	-0.061	0.052
50%	LLM	0.007	0.017	0.004	-0.019	0.036
50%	Human Validation	-0.004	0.013	-0.002	-0.023	0.016
50%	Debiased	0.000	0.035	0.000	-0.054	0.057

Table 13: Bias Summary Statistics of β_{20} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	-0.001	0.018	-0.002	-0.036	0.025
5%	Human Validation	-0.010	0.058	-0.003	-0.111	0.090
5%	Debiased	-0.012	0.064	-0.002	-0.102	0.088
10%	LLM	-0.001	0.018	0.000	-0.033	0.025
10%	Human Validation	0.000	0.033	-0.002	-0.047	0.058
10%	Debiased	-0.002	0.039	0.001	-0.068	0.051
25%	LLM	0.000	0.019	0.000	-0.029	0.024
25%	Human Validation	0.000	0.020	0.000	-0.030	0.036
25%	Debiased	-0.004	0.030	-0.001	-0.049	0.039
50%	LLM	-0.003	0.019	0.000	-0.032	0.025
50%	Human Validation	-0.002	0.018	0.000	-0.032	0.025
50%	Debiased	-0.002	0.026	0.000	-0.048	0.043

Table 14: Bias Summary Statistics of β_{Other} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Bias	Mean	SD	Median	5%	95%
5%	LLM	0.000	0.010	-0.001	-0.016	0.016
5%	Human Validation	-0.002	0.023	-0.001	-0.038	0.039
5%	Debiased	-0.031	0.244	-0.001	-0.072	0.058
10%	LLM	0.000	0.009	0.000	-0.015	0.014
10%	Human Validation	-0.002	0.019	-0.001	-0.032	0.027
10%	Debiased	0.001	0.023	0.000	-0.037	0.032
25%	LLM	0.000	0.010	0.000	-0.014	0.018
25%	Human Validation	0.000	0.015	0.000	-0.027	0.024
25%	Debiased	-0.001	0.016	-0.001	-0.027	0.028
50%	LLM	-0.001	0.009	-0.001	-0.016	0.015
50%	Human Validation	0.000	0.009	0.000	-0.013	0.013
50%	Debiased	0.003	0.016	0.001	-0.026	0.027

Table 15: Normalized Bias Summary Statistics of β_3 by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	0.083	12.266	0.088	-3.428	5.991
5%	Human Validation	-0.680	8.142	0.115	-2.922	3.310
5%	Debiased	0.370	3.064	0.008	-1.517	2.169
10%	LLM	-7.306	70.594	0.386	-4.590	3.767
10%	Human Validation	-0.126	2.849	-0.056	-2.549	2.947
10%	Debiased	-0.400	2.651	-0.009	-3.940	1.964
25%	LLM	0.981	5.104	0.202	-1.928	3.935
25%	Human Validation	1.966	15.681	-0.005	-5.196	2.876
25%	Debiased	1.340	7.448	-0.012	-3.773	7.065
50%	LLM	-1.784	12.476	0.176	-5.591	2.737
50%	Human Validation	0.111	3.649	-0.050	-2.369	3.386
50%	Debiased	-1.773	12.795	-0.001	-9.301	4.413

Table 16: Normalized Bias Summary Statistics of β_{14} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	-1.549300e+01	1.308840e+02	-0.392	-5.006	3.532
5%	Human Validation	-2.332542e+13	2.430916e+14	-0.048	-5.051	3.928
5%	Debiased	1.690000e-01	2.867000e+00	0.125	-4.029	2.491
10%	LLM	8.420000e-01	1.192900e+01	-0.631	-10.345	9.609
10%	Human Validation	Inf	NaN	-0.019	-2.704	4.710
10%	Debiased	-1.535000e+00	1.574700e+01	-0.028	-6.187	6.686
25%	LLM	-1.957000e+00	1.344400e+01	-0.395	-13.683	6.068
25%	Human Validation	3.040000e-01	5.447000e+00	0.043	-3.155	5.194
25%	Debiased	-4.420000e-01	2.829000e+00	-0.011	-6.011	2.548
50%	LLM	6.042000e+00	6.228400e+01	-0.441	-6.986	11.804
50%	Human Validation	-2.690000e-01	2.475000e+00	0.042	-3.805	2.496
50%	Debiased	3.230000e-01	3.471000e+00	-0.028	-2.653	3.096

Table 17: Normalized Bias Summary Statistics of β_{15} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	-1.261	8.078	0.145	-6.557	2.313
5%	Human Validation	0.150	3.947	-0.028	-4.559	5.185
5%	Debiased	0.275	2.904	0.022	-2.589	3.520
10%	LLM	0.956	14.402	0.248	-6.783	3.603
10%	Human Validation	0.399	7.416	-0.111	-4.292	2.558
10%	Debiased	-0.121	3.433	-0.014	-2.963	2.961
25%	LLM	-2.758	18.172	-0.200	-4.509	2.257
25%	Human Validation	-1.175	5.297	-0.183	-7.113	3.772
25%	Debiased	0.301	3.150	0.017	-2.744	3.738
50%	LLM	-0.078	6.440	0.273	-5.370	4.930
50%	Human Validation	0.377	5.840	0.235	-3.729	6.951
50%	Debiased	-0.087	2.329	-0.043	-3.257	2.435

Table 18: Normalized Bias Summary Statistics of β_{19} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	1.975333e+12	1.676126e+13	0.325	-7.525	6.716
5%	Human Validation	-6.400000e-01	3.075000e+00	-0.018	-5.918	1.724
5%	Debiased	1.346000e+00	1.080000e+01	-0.012	-1.748	3.105
10%	LLM	1.747000e+00	1.305300e+01	0.448	-2.519	16.339
10%	Human Validation	7.960000e-01	7.585000e+00	0.228	-3.007	3.838
10%	Debiased	2.730000e-01	2.410000e+00	0.072	-1.963	2.396
25%	LLM	-1.990000e-01	1.291300e+01	0.388	-5.285	7.043
25%	Human Validation	-2.410000e-01	4.783000e+00	0.002	-1.848	2.915
25%	Debiased	1.950000e-01	9.399000e+00	0.030	-2.975	1.895
50%	LLM	1.031000e+00	1.059500e+01	0.345	-3.221	6.766
50%	Human Validation	1.800000e-02	2.836000e+00	-0.136	-2.395	3.319
50%	Debiased	-7.310000e-01	4.016000e+00	-0.005	-8.124	1.229

Table 19: Normalized Bias Summary Statistics of β_{20} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	-6.760000e-01	5.915000e+00	-0.130	-4.650	4.556
5%	Human Validation	NaN	NaN	-0.041	-10.759	2.346
5%	Debiased	-1.188300e+01	1.024120e+02	-0.021	-3.760	5.343
10%	LLM	-2.470000e+00	2.543100e+01	0.030	-2.269	4.469
10%	Human Validation	-4.279744e+11	3.631483e+12	-0.030	-3.323	2.177
10%	Debiased	3.118000e+00	2.039500e+01	0.024	-2.439	3.886
25%	LLM	-1.890000e-01	5.584000e+00	-0.002	-3.308	4.861
25%	Human Validation	3.040000e-01	3.924000e+00	-0.010	-2.259	5.698
25%	Debiased	-1.720000e-01	1.742000e+00	-0.020	-2.044	2.017
50%	LLM	-7.180000e-01	3.407000e+00	-0.046	-6.274	2.526
50%	Human Validation	2.173257e+11	1.844070e+12	0.010	-7.188	2.020
50%	Debiased	-3.640000e-01	2.172000e+00	-0.003	-3.959	2.441

Table 20: Normalized Bias Summary Statistics of β_{Other} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
5%	LLM	-2.004	13.984	-0.085	-5.244	3.555
5%	Human Validation	-0.670	4.767	-0.014	-6.360	2.658
5%	Debiased	0.092	2.699	-0.031	-2.068	4.828
10%	LLM	7.598	131.799	-0.030	-15.531	9.111
10%	Human Validation	-0.801	6.152	-0.028	-4.520	3.526
10%	Debiased	0.518	5.145	-0.011	-2.939	2.666
25%	LLM	-0.254	36.374	0.053	-2.854	15.543
25%	Human Validation	0.476	6.166	0.013	-4.100	4.887
25%	Debiased	-0.231	7.164	-0.071	-4.973	2.920
50%	LLM	-1.052	5.492	-0.201	-11.533	4.224
50%	Human Validation	0.222	3.834	0.015	-2.630	2.549
50%	Debiased	-0.042	2.935	0.023	-2.854	2.568

Table 21: Bias Summary Statistics of β_3 by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.005	0.032	0.012	-0.045	0.053
GPT-3.5	5%	Human Validation	0.027	0.074	0.023	-0.078	0.130
GPT-3.5	5%	Debiased	-0.006	0.104	0.000	-0.234	0.136
GPT-3.5	10%	LLM	0.005	0.033	0.009	-0.047	0.046
GPT-3.5	10%	Human Validation	-0.005	0.054	-0.006	-0.085	0.098
GPT-3.5	10%	Debiased	-0.002	0.081	0.002	-0.125	0.139
GPT-3.5	25%	LLM	0.005	0.034	0.008	-0.054	0.057
GPT-3.5	25%	Human Validation	0.006	0.029	0.003	-0.039	0.044
GPT-3.5	25%	Debiased	0.000	0.044	0.001	-0.067	0.072
GPT-3.5	50%	LLM	0.000	0.031	0.004	-0.049	0.044
GPT-3.5	50%	Human Validation	-0.002	0.020	-0.001	-0.045	0.028
GPT-3.5	50%	Debiased	-0.009	0.041	-0.006	-0.090	0.061
GPT-4o	5%	LLM	-0.001	0.018	-0.003	-0.027	0.027
GPT-4o	5%	Human Validation	-0.003	0.074	-0.001	-0.107	0.102
GPT-4o	5%	Debiased	-0.008	0.058	0.000	-0.115	0.062
GPT-4o	10%	LLM	0.006	0.016	0.005	-0.016	0.033
GPT-4o	10%	Human Validation	0.007	0.053	0.000	-0.066	0.092
GPT-4o	10%	Debiased	0.003	0.048	-0.001	-0.068	0.081
GPT-4o	25%	LLM	0.003	0.016	0.001	-0.021	0.024
GPT-4o	25%	Human Validation	-0.007	0.025	-0.007	-0.046	0.032
GPT-4o	25%	Debiased	0.000	0.035	-0.001	-0.048	0.056
GPT-4o	50%	LLM	0.001	0.016	0.001	-0.025	0.025
GPT-4o	50%	Human Validation	-0.002	0.019	-0.003	-0.032	0.030
GPT-4o	50%	Debiased	0.003	0.036	0.002	-0.064	0.056

Table 22: Bias Summary Statistics of β_{14} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^{\top}\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.000	0.039	-0.010	-0.049	0.063
GPT-3.5	5%	Human Validation	-0.024	0.096	-0.032	-0.149	0.152
GPT-3.5	5%	Debiased	0.723	4.079	0.049	-0.245	0.512
GPT-3.5	10%	LLM	-0.003	0.039	-0.015	-0.050	0.056
GPT-3.5	10%	Human Validation	-0.008	0.047	-0.003	-0.082	0.072
GPT-3.5	10%	Debiased	-0.032	0.141	-0.036	-0.246	0.184
GPT-3.5	25%	LLM	-0.006	0.040	-0.016	-0.062	0.060
GPT-3.5	25%	Human Validation	0.002	0.032	0.002	-0.039	0.060
GPT-3.5	25%	Debiased	-0.016	0.096	-0.008	-0.167	0.114
GPT-3.5	50%	LLM	-0.007	0.042	-0.019	-0.060	0.056
GPT-3.5	50%	Human Validation	-0.004	0.023	0.001	-0.045	0.030
GPT-3.5	50%	Debiased	-0.002	0.096	0.002	-0.137	0.192
GPT-4o	5%	LLM	-0.001	0.030	-0.006	-0.046	0.043
GPT-4o	5%	Human Validation	-0.005	0.078	-0.002	-0.142	0.107
GPT-4o	5%	Debiased	0.020	0.192	0.020	-0.244	0.330
GPT-4o	10%	LLM	0.002	0.029	-0.006	-0.035	0.055
GPT-4o	10%	Human Validation	-0.002	0.072	-0.001	-0.132	0.111
GPT-4o	10%	Debiased	0.004	0.105	-0.005	-0.152	0.189
GPT-4o	25%	LLM	-0.002	0.029	-0.003	-0.044	0.041
GPT-4o	25%	Human Validation	0.005	0.040	0.002	-0.049	0.077
GPT-4o	25%	Debiased	0.001	0.063	0.001	-0.084	0.090
GPT-4o	50%	LLM	0.000	0.028	-0.005	-0.038	0.040
GPT-4o	50%	Human Validation	-0.004	0.025	0.002	-0.045	0.030
GPT-4o	50%	Debiased	-0.016	0.057	-0.007	-0.100	0.086

Table 23: Bias Summary Statistics of β_{15} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.001	0.017	-0.002	-0.023	0.031
GPT-3.5	5%	Human Validation	0.002	0.061	-0.008	-0.074	0.111
GPT-3.5	5%	Debiased	0.010	0.116	0.004	-0.174	0.166
GPT-3.5	10%	LLM	0.002	0.024	0.003	-0.032	0.051
GPT-3.5	10%	Human Validation	-0.014	0.044	-0.024	-0.084	0.057
GPT-3.5	10%	Debiased	0.007	0.042	0.001	-0.053	0.079
GPT-3.5	25%	LLM	-0.004	0.022	-0.004	-0.039	0.029
GPT-3.5	25%	Human Validation	-0.004	0.025	-0.004	-0.038	0.038
GPT-3.5	25%	Debiased	0.001	0.051	-0.002	-0.071	0.075
GPT-3.5	50%	LLM	0.004	0.016	0.001	-0.020	0.031
GPT-3.5	50%	Human Validation	0.004	0.019	0.003	-0.022	0.040
GPT-3.5	50%	Debiased	-0.014	0.040	-0.005	-0.077	0.046
GPT-4o	5%	LLM	0.003	0.018	0.003	-0.025	0.033
GPT-4o	5%	Human Validation	-0.015	0.058	-0.001	-0.103	0.061
GPT-4o	5%	Debiased	-0.005	0.083	0.002	-0.138	0.131
GPT-4o	10%	LLM	0.008	0.020	0.008	-0.020	0.038
GPT-4o	10%	Human Validation	-0.006	0.037	-0.002	-0.067	0.044
GPT-4o	10%	Debiased	-0.005	0.059	-0.002	-0.123	0.090
GPT-4o	25%	LLM	0.001	0.018	-0.001	-0.021	0.032
GPT-4o	25%	Human Validation	-0.008	0.027	-0.005	-0.055	0.031
GPT-4o	25%	Debiased	0.011	0.043	0.004	-0.051	0.071
GPT-4o	50%	LLM	0.013	0.017	0.011	-0.014	0.045
GPT-4o	50%	Human Validation	0.007	0.017	0.007	-0.020	0.029
GPT-4o	50%	Debiased	0.004	0.041	0.000	-0.047	0.081

Table 24: Bias Summary Statistics of β_{19} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.009	0.022	0.014	-0.033	0.037
GPT-3.5	5%	Human Validation	-0.001	0.046	-0.001	-0.064	0.091
GPT-3.5	5%	Debiased	0.039	0.180	0.001	-0.091	0.197
GPT-3.5	10%	LLM	0.006	0.017	0.010	-0.021	0.028
GPT-3.5	10%	Human Validation	0.008	0.041	0.008	-0.063	0.067
GPT-3.5	10%	Debiased	0.001	0.065	0.002	-0.101	0.101
GPT-3.5	25%	LLM	0.004	0.019	0.003	-0.029	0.031
GPT-3.5	25%	Human Validation	-0.002	0.018	0.000	-0.028	0.030
GPT-3.5	25%	Debiased	0.008	0.034	0.007	-0.049	0.064
GPT-3.5	50%	LLM	0.009	0.021	0.006	-0.021	0.040
GPT-3.5	50%	Human Validation	-0.003	0.012	-0.001	-0.020	0.016
GPT-3.5	50%	Debiased	0.005	0.035	0.006	-0.055	0.056
GPT-4o	5%	LLM	0.001	0.014	-0.001	-0.016	0.026
GPT-4o	5%	Human Validation	-0.018	0.052	-0.004	-0.110	0.061
GPT-4o	5%	Debiased	0.003	0.062	-0.003	-0.103	0.108
GPT-4o	10%	LLM	0.004	0.016	0.007	-0.019	0.024
GPT-4o	10%	Human Validation	0.006	0.038	0.006	-0.053	0.057
GPT-4o	10%	Debiased	0.007	0.050	0.018	-0.083	0.085
GPT-4o	25%	LLM	0.004	0.012	0.004	-0.012	0.023
GPT-4o	25%	Human Validation	0.001	0.017	0.000	-0.025	0.032
GPT-4o	25%	Debiased	-0.001	0.034	0.000	-0.061	0.038
GPT-4o	50%	LLM	0.005	0.011	0.004	-0.010	0.021
GPT-4o	50%	Human Validation	-0.004	0.014	-0.003	-0.029	0.014
GPT-4o	50%	Debiased	-0.005	0.034	-0.006	-0.047	0.053

Table 25: Bias Summary Statistics of β_{20} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.000	0.018	-0.002	-0.032	0.026
GPT-3.5	5%	Human Validation	-0.004	0.058	-0.001	-0.105	0.078
GPT-3.5	5%	Debiased	-0.016	0.061	-0.005	-0.097	0.073
GPT-3.5	10%	LLM	0.002	0.018	0.004	-0.025	0.028
GPT-3.5	10%	Human Validation	-0.001	0.036	-0.002	-0.052	0.060
GPT-3.5	10%	Debiased	-0.002	0.037	0.001	-0.067	0.045
GPT-3.5	25%	LLM	0.003	0.018	0.004	-0.023	0.023
GPT-3.5	25%	Human Validation	0.000	0.020	0.001	-0.029	0.031
GPT-3.5	25%	Debiased	-0.005	0.034	-0.005	-0.053	0.045
GPT-3.5	50%	LLM	-0.001	0.017	0.000	-0.030	0.027
GPT-3.5	50%	Human Validation	-0.005	0.019	0.000	-0.044	0.019
GPT-3.5	50%	Debiased	-0.003	0.026	-0.001	-0.043	0.040
GPT-4o	5%	LLM	-0.002	0.019	-0.002	-0.037	0.025
GPT-4o	5%	Human Validation	-0.016	0.058	-0.013	-0.117	0.099
GPT-4o	5%	Debiased	-0.009	0.067	-0.002	-0.120	0.085
GPT-4o	10%	LLM	-0.004	0.019	-0.002	-0.035	0.021
GPT-4o	10%	Human Validation	0.001	0.030	-0.001	-0.040	0.051
GPT-4o	10%	Debiased	-0.003	0.042	0.001	-0.068	0.059
GPT-4o	25%	LLM	-0.004	0.019	-0.002	-0.038	0.023
GPT-4o	25%	Human Validation	0.000	0.020	-0.001	-0.027	0.042
GPT-4o	25%	Debiased	-0.004	0.025	0.000	-0.046	0.032
GPT-4o	50%	LLM	-0.005	0.020	-0.002	-0.039	0.019
GPT-4o	50%	Human Validation	0.002	0.017	0.001	-0.028	0.030
GPT-4o	50%	Debiased	-0.002	0.027	0.001	-0.048	0.046

Table 26: Bias Summary Statistics of β_{Other} by Model and Proportion of Validation Samples for. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.000	0.012	0.000	-0.017	0.018
GPT-3.5	5%	Human Validation	-0.001	0.027	0.001	-0.045	0.043
GPT-3.5	5%	Debiased	-0.063	0.343	-0.003	-0.082	0.052
GPT-3.5	10%	LLM	0.000	0.011	0.000	-0.016	0.017
GPT-3.5	10%	Human Validation	-0.004	0.019	-0.002	-0.030	0.028
GPT-3.5	10%	Debiased	0.003	0.026	0.000	-0.039	0.035
GPT-3.5	25%	LLM	-0.001	0.011	-0.002	-0.018	0.019
GPT-3.5	25%	Human Validation	-0.002	0.015	0.000	-0.030	0.023
GPT-3.5	25%	Debiased	-0.001	0.017	-0.002	-0.025	0.028
GPT-3.5	50%	LLM	-0.001	0.010	-0.001	-0.016	0.015
GPT-3.5	50%	Human Validation	0.000	0.008	0.000	-0.011	0.013
GPT-3.5	50%	Debiased	0.001	0.016	0.000	-0.022	0.036
GPT-4o	5%	LLM	0.000	0.007	-0.001	-0.010	0.012
GPT-4o	5%	Human Validation	-0.004	0.019	-0.004	-0.035	0.029
GPT-4o	5%	Debiased	0.001	0.032	0.001	-0.042	0.065
GPT-4o	10%	LLM	-0.001	0.007	0.000	-0.011	0.008
GPT-4o	10%	Human Validation	-0.001	0.019	0.000	-0.033	0.027
GPT-4o	10%	Debiased	-0.001	0.020	0.000	-0.035	0.023
GPT-4o	25%	LLM	0.000	0.008	0.001	-0.011	0.013
GPT-4o	25%	Human Validation	0.002	0.015	0.002	-0.025	0.024
GPT-4o	25%	Debiased	-0.001	0.016	-0.001	-0.027	0.027
GPT-4o	50%	LLM	-0.001	0.008	-0.001	-0.014	0.013
GPT-4o	50%	Human Validation	-0.001	0.010	0.000	-0.016	0.014
GPT-4o	50%	Debiased	0.004	0.015	0.003	-0.027	0.024

Table 27: Normalized Bias Summary of β_3 Statistics by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.404	17.377	0.798	-5.028	16.161
GPT-3.5	5%	Human Validation	0.784	2.391	0.315	-1.177	3.128
GPT-3.5	5%	Debiased	0.736	4.131	-0.001	-1.593	3.369
GPT-3.5	10%	LLM	-14.808	99.951	0.376	-6.857	5.317
GPT-3.5	10%	Human Validation	-0.061	1.932	-0.110	-2.151	1.928
GPT-3.5	10%	Debiased	-0.750	3.483	0.018	-5.075	1.653
GPT-3.5	25%	LLM	0.700	2.141	0.366	-1.916	4.716
GPT-3.5	25%	Human Validation	3.388	19.759	0.067	-2.092	2.346
GPT-3.5	25%	Debiased	1.547	8.455	0.013	-3.892	8.236
GPT-3.5	50%	LLM	-2.161	15.770	0.212	-6.645	4.589
GPT-3.5	50%	Human Validation	-0.433	3.036	-0.032	-3.063	2.304
GPT-3.5	50%	Debiased	-3.477	17.859	-0.197	-12.129	4.605
GPT-4o	5%	LLM	-0.238	1.735	-0.141	-2.654	2.305
GPT-4o	5%	Human Validation	-2.144	11.152	-0.008	-9.991	3.172
GPT-4o	5%	Debiased	0.003	1.304	0.015	-1.315	1.818
GPT-4o	10%	LLM	0.196	1.851	0.386	-2.094	1.863
GPT-4o	10%	Human Validation	-0.190	3.566	0.000	-3.104	3.606
GPT-4o	10%	Debiased	-0.051	1.368	-0.017	-2.609	2.363
GPT-4o	25%	LLM	1.262	6.935	0.033	-1.395	2.564
GPT-4o	25%	Human Validation	0.544	10.208	-0.272	-9.297	9.053
GPT-4o	25%	Debiased	1.133	6.398	-0.022	-3.360	6.854
GPT-4o	50%	LLM	-1.407	8.170	0.106	-4.840	2.473
GPT-4o	50%	Human Validation	0.655	4.146	-0.077	-1.779	3.586
GPT-4o	50%	Debiased	-0.068	2.681	0.053	-6.847	3.269

Table 28: Normalized Bias Summary Statistics of β_{14} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	-3.109500e+01	1.850530e+02	-0.349	-6.156	3.647
GPT-3.5	5%	Human Validation	-4.665084e+13	3.446102e+14	-0.290	-6.090	14.412
GPT-3.5	5%	Debiased	7.860000e-01	3.240000e+00	0.152	-3.387	5.784
GPT-3.5	10%	LLM	1.473000e+00	1.569700e+01	-0.746	-11.986	11.050
GPT-3.5	10%	Human Validation	Inf	NaN	-0.040	-2.126	39.905
GPT-3.5	10%	Debiased	-5.120000e-01	4.900000e+00	-0.266	-4.905	6.427
GPT-3.5	25%	LLM	-9.260000e-01	1.173100e+01	-0.757	-4.610	7.003
GPT-3.5	25%	Human Validation	-3.300000e-02	2.732000e+00	0.043	-4.256	3.281
GPT-3.5	25%	Debiased	-2.160000e-01	1.369000e+00	-0.085	-2.409	2.004
GPT-3.5	50%	LLM	9.614000e+00	8.732900e+01	-1.021	-21.102	21.792
GPT-3.5	50%	Human Validation	-3.330000e-01	2.836000e+00	0.035	-4.525	2.742
GPT-3.5	50%	Debiased	-3.060000e-01	1.841000e+00	0.014	-3.508	2.772
GPT-4o	5%	LLM	1.090000e-01	2.264000e+00	-0.397	-2.306	2.945
GPT-4o	5%	Human Validation	-1.300000e-02	1.634000e+00	-0.034	-2.172	3.481
GPT-4o	5%	Debiased	-4.480000e-01	2.321000e+00	0.122	-5.526	1.910
GPT-4o	10%	LLM	2.120000e-01	6.439000e+00	-0.543	-6.251	8.938
GPT-4o	10%	Human Validation	-6.600000e-02	2.126000e+00	-0.011	-2.672	4.209
GPT-4o	10%	Debiased	-2.559000e+00	2.183700e+01	-0.027	-6.992	4.811
GPT-4o	25%	LLM	-2.988000e+00	1.506100e+01	-0.105	-16.425	5.192
GPT-4o	25%	Human Validation	6.400000e-01	7.245000e+00	0.027	-2.297	5.921
GPT-4o	25%	Debiased	-6.690000e-01	3.775000e+00	0.011	-9.549	2.573
GPT-4o	50%	LLM	2.471000e+00	1.472800e+01	-0.218	-3.930	6.686
GPT-4o	50%	Human Validation	-2.060000e-01	2.091000e+00	0.047	-3.273	2.259
GPT-4o	50%	Debiased	9.530000e-01	4.499000e+00	-0.081	-2.059	11.398

Table 29: Normalized Bias Summary Statistics of β_{15} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	-1.302	5.770	-0.080	-9.616	2.015
GPT-3.5	5%	Human Validation	0.777	5.259	-0.065	-4.125	9.894
GPT-3.5	5%	Debiased	0.085	2.198	0.021	-3.006	1.911
GPT-3.5	10%	LLM	-1.505	7.424	0.120	-10.828	3.269
GPT-3.5	10%	Human Validation	1.128	10.306	-0.437	-5.260	4.922
GPT-3.5	10%	Debiased	0.571	2.443	0.011	-1.671	5.167
GPT-3.5	25%	LLM	-1.632	6.161	-0.415	-5.893	1.355
GPT-3.5	25%	Human Validation	-1.886	6.738	-0.161	-7.331	2.157
GPT-3.5	25%	Debiased	0.226	3.709	-0.027	-2.557	2.415
GPT-3.5	50%	LLM	-0.912	8.731	0.068	-7.264	5.061
GPT-3.5	50%	Human Validation	0.140	3.266	0.201	-2.846	3.475
GPT-3.5	50%	Debiased	-0.246	1.245	-0.204	-1.991	2.397
GPT-4o	5%	LLM	-1.220	9.953	0.167	-2.346	2.715
GPT-4o	5%	Human Validation	-0.476	1.768	-0.014	-3.113	1.191
GPT-4o	5%	Debiased	0.466	3.493	0.031	-1.713	4.039
GPT-4o	10%	LLM	3.417	18.793	0.336	-5.509	7.626
GPT-4o	10%	Human Validation	-0.331	2.066	-0.028	-2.732	2.298
GPT-4o	10%	Debiased	-0.814	4.118	-0.047	-3.262	1.519
GPT-4o	25%	LLM	-3.884	25.087	-0.025	-1.845	2.922
GPT-4o	25%	Human Validation	-0.464	3.239	-0.183	-6.686	5.972
GPT-4o	25%	Debiased	0.376	2.522	0.090	-2.263	3.746
GPT-4o	50%	LLM	0.757	2.543	0.430	-3.431	4.380
GPT-4o	50%	Human Validation	0.613	7.642	0.303	-3.774	13.178
GPT-4o	50%	Debiased	0.073	3.066	0.006	-4.922	3.059

Table 30: Normalized Bias Summary Statistics of β_{19} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	3.950666e+12	2.3704e+13	1.003	-8.309	7.904
GPT-3.5	5%	Human Validation	-4.500000e-01	3.6550e+00	-0.018	-6.221	2.703
GPT-3.5	5%	Debiased	2.250000e+00	1.5122e+01	0.007	-2.092	1.268
GPT-3.5	10%	LLM	3.691000e+00	1.4338e+01	0.482	-1.427	17.183
GPT-3.5	10%	Human Validation	2.138000e+00	9.1650e+00	0.274	-1.701	13.084
GPT-3.5	10%	Debiased	3.070000e-01	3.1620e+00	0.022	-1.514	1.204
GPT-3.5	25%	LLM	-1.518000e+00	1.7932e+01	0.435	-12.699	10.846
GPT-3.5	25%	Human Validation	-8.550000e-01	6.2000e+00	0.002	-1.687	2.905
GPT-3.5	25%	Debiased	-5.640000e-01	5.1740e+00	0.080	-2.255	1.963
GPT-3.5	50%	LLM	1.638000e+00	1.4927e+01	0.423	-3.093	11.338
GPT-3.5	50%	Human Validation	-2.240000e-01	2.2760e+00	-0.086	-2.897	1.457
GPT-3.5	50%	Debiased	2.620000e-01	2.5980e+00	0.111	-1.895	2.350
GPT-4o	5%	LLM	-3.320000e-01	5.6010e+00	-0.100	-5.390	3.583
GPT-4o	5%	Human Validation	-8.300000e-01	2.3980e+00	-0.071	-4.661	1.497
GPT-4o	5%	Debiased	4.410000e-01	2.4990e+00	-0.021	-1.603	5.011
GPT-4o	10%	LLM	-1.980000e-01	1.1500e+01	0.365	-4.053	10.623
GPT-4o	10%	Human Validation	-5.450000e-01	5.3850e+00	0.179	-3.071	2.036
GPT-4o	10%	Debiased	2.400000e-01	1.3330e+00	0.227	-1.955	2.832
GPT-4o	25%	LLM	1.120000e+00	3.6270e+00	0.327	-1.055	4.889
GPT-4o	25%	Human Validation	3.720000e-01	2.6840e+00	-0.006	-1.636	2.724
GPT-4o	25%	Debiased	9.530000e-01	1.2298e+01	-0.001	-3.690	1.557
GPT-4o	50%	LLM	4.240000e-01	2.0350e+00	0.299	-3.010	2.898
GPT-4o	50%	Human Validation	2.610000e-01	3.3190e+00	-0.207	-1.972	4.315
GPT-4o	50%	Debiased	-1.723000e+00	4.8930e+00	-0.187	-12.550	1.052

Table 31: Normalized Bias Summary Statistics of β_{20} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	-9.500000e-02	3.691000e+00	-0.063	-2.512	3.015
GPT-3.5	5%	Human Validation	Inf	NaN	-0.008	-3.529	Inf
GPT-3.5	5%	Debiased	1.880000e-01	2.184000e+00	-0.045	-2.232	5.210
GPT-3.5	10%	LLM	-5.666000e+00	3.556000e+01	0.135	-2.986	2.802
GPT-3.5	10%	Human Validation	-6.210000e-01	4.357000e+00	-0.037	-3.229	1.979
GPT-3.5	10%	Debiased	6.237000e+00	2.863700e+01	0.056	-2.509	21.176
GPT-3.5	25%	LLM	9.070000e-01	4.613000e+00	0.303	-2.927	6.805
GPT-3.5	25%	Human Validation	-1.440000e-01	3.979000e+00	0.021	-3.348	5.197
GPT-3.5	25%	Debiased	-2.840000e-01	1.428000e+00	-0.075	-2.424	1.989
GPT-3.5	50%	LLM	-8.360000e-01	3.876000e+00	0.007	-6.841	1.871
GPT-3.5	50%	Human Validation	-2.938000e+00	1.362700e+01	-0.015	-8.076	1.806
GPT-3.5	50%	Debiased	-5.070000e-01	2.718000e+00	-0.020	-4.304	3.035
GPT-4o	5%	LLM	-1.258000e+00	7.527000e+00	-0.160	-5.650	4.265
GPT-4o	5%	Human Validation	-Inf	NaN	-0.252	-15.255	1.615
GPT-4o	5%	Debiased	-2.395500e+01	1.448150e+02	-0.021	-4.945	3.579
GPT-4o	10%	LLM	7.260000e-01	5.142000e+00	-0.162	-1.716	5.355
GPT-4o	10%	Human Validation	-8.559488e+11	5.135693e+12	-0.016	-3.251	2.349
GPT-4o	10%	Debiased	0.000000e+00	1.937000e+00	0.022	-2.038	2.651
GPT-4o	25%	LLM	-1.285000e+00	6.285000e+00	-0.094	-6.183	2.126
GPT-4o	25%	Human Validation	7.510000e-01	3.873000e+00	-0.021	-1.570	5.652
GPT-4o	25%	Debiased	-5.900000e-02	2.023000e+00	0.009	-1.177	1.481
GPT-4o	50%	LLM	-6.010000e-01	2.914000e+00	-0.206	-5.165	3.387
GPT-4o	50%	Human Validation	4.346514e+11	2.607908e+12	0.051	-1.711	27.212
GPT-4o	50%	Debiased	-2.220000e-01	1.463000e+00	0.017	-2.596	1.966

Table 32: Normalized Bias Summary Statistics of β_{Other} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	Normalized Bias	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	-3.191	17.243	-0.048	-6.641	2.855
GPT-3.5	5%	Human Validation	-0.142	4.516	0.014	-6.222	4.212
GPT-3.5	5%	Debiased	0.024	1.182	-0.077	-1.314	2.036
GPT-3.5	10%	LLM	27.661	171.794	-0.018	-11.593	8.068
GPT-3.5	10%	Human Validation	-2.019	7.410	-0.196	-8.176	2.063
GPT-3.5	10%	Debiased	1.096	7.029	-0.011	-3.734	6.555
GPT-3.5	25%	LLM	-4.367	45.766	-0.215	-4.979	21.277
GPT-3.5	25%	Human Validation	-0.624	3.536	-0.018	-5.422	2.497
GPT-3.5	25%	Debiased	0.927	7.698	-0.084	-5.179	6.291
GPT-3.5	50%	LLM	-0.727	3.699	-0.129	-6.517	4.207
GPT-3.5	50%	Human Validation	-0.002	3.501	0.015	-3.252	3.870
GPT-3.5	50%	Debiased	-0.195	2.589	-0.008	-3.833	2.414
GPT-4o	5%	LLM	-0.817	9.821	-0.121	-2.226	4.974
GPT-4o	5%	Human Validation	-1.198	5.012	-0.133	-5.746	1.786
GPT-4o	5%	Debiased	0.161	3.657	0.029	-3.823	5.490
GPT-4o	10%	LLM	-12.465	69.978	-0.077	-25.423	9.083
GPT-4o	10%	Human Validation	0.417	4.337	-0.001	-3.938	5.422
GPT-4o	10%	Debiased	-0.060	1.902	0.006	-1.910	2.382
GPT-4o	25%	LLM	3.860	23.551	0.186	-2.380	2.798
GPT-4o	25%	Human Validation	1.575	7.883	0.120	-2.376	5.889
GPT-4o	25%	Debiased	-1.390	6.487	-0.050	-4.256	2.677
GPT-4o	50%	LLM	-1.378	6.876	-0.308	-13.354	2.533
GPT-4o	50%	Human Validation	0.447	4.179	0.014	-1.860	1.821
GPT-4o	50%	Debiased	0.110	3.275	0.198	-2.526	2.520

Table 33: MSE Summary Statistics of β_3 by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.001	0.001	0.000	0.003
5%	Human Validation	0.012	0.012	0.009	0.001	0.031
5%	Debiased	0.016	0.025	0.008	0.001	0.060
10%	LLM	0.001	0.001	0.001	0.000	0.003
10%	Human Validation	0.007	0.007	0.006	0.001	0.021
10%	Debiased	0.009	0.011	0.005	0.000	0.030
25%	LLM	0.001	0.001	0.001	0.000	0.004
25%	Human Validation	0.002	0.002	0.002	0.000	0.005
25%	Debiased	0.003	0.003	0.002	0.000	0.008
50%	LLM	0.001	0.001	0.001	0.000	0.003
50%	Human Validation	0.001	0.001	0.001	0.000	0.002
50%	Debiased	0.004	0.004	0.003	0.000	0.012

Table 34: MSE Summary Statistics of β_{14} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.001	0.001	0.000	0.004
5%	Human Validation	0.017	0.017	0.011	0.002	0.048
5%	Debiased	16.716	140.921	0.048	0.005	0.547
10%	LLM	0.001	0.001	0.001	0.000	0.003
10%	Human Validation	0.007	0.007	0.005	0.000	0.024
10%	Debiased	0.033	0.042	0.025	0.003	0.111
25%	LLM	0.001	0.002	0.001	0.000	0.004
25%	Human Validation	0.002	0.002	0.002	0.000	0.007
25%	Debiased	0.016	0.017	0.013	0.001	0.047
50%	LLM	0.001	0.001	0.001	0.000	0.004
50%	Human Validation	0.001	0.001	0.001	0.000	0.003
50%	Debiased	0.014	0.013	0.011	0.001	0.043

Table 35: MSE Summary Statistics of β_{15} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.000	0.000	0	0.002
5%	Human Validation	0.008	0.009	0.005	0	0.026
5%	Debiased	0.019	0.022	0.013	0	0.058
10%	LLM	0.001	0.001	0.001	0	0.003
10%	Human Validation	0.003	0.003	0.003	0	0.008
10%	Debiased	0.007	0.007	0.005	0	0.023
25%	LLM	0.001	0.001	0.000	0	0.002
25%	Human Validation	0.002	0.002	0.001	0	0.004
25%	Debiased	0.005	0.005	0.004	0	0.012
50%	LLM	0.001	0.001	0.000	0	0.002
50%	Human Validation	0.001	0.001	0.001	0	0.002
50%	Debiased	0.004	0.004	0.003	0	0.011

Table 36: MSE Summary Statistics of β_{19} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.001	0.000	0.000	0.002
5%	Human Validation	0.005	0.004	0.004	0.000	0.014
5%	Debiased	0.036	0.189	0.011	0.001	0.041
10%	LLM	0.000	0.001	0.000	0.000	0.001
10%	Human Validation	0.003	0.003	0.003	0.000	0.008
10%	Debiased	0.008	0.008	0.006	0.001	0.023
25%	LLM	0.000	0.000	0.000	0.000	0.001
25%	Human Validation	0.001	0.001	0.001	0.000	0.002
25%	Debiased	0.004	0.005	0.002	0.000	0.015
50%	LLM	0.000	0.001	0.000	0.000	0.002
50%	Human Validation	0.000	0.000	0.000	0.000	0.001
50%	Debiased	0.003	0.002	0.002	0.000	0.008

Table 37: MSE Summary Statistics of β_{20} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.001	0.001	0.000	0.000	0.002
5%	Human Validation	0.007	0.006	0.005	0.000	0.017
5%	Debiased	0.010	0.009	0.009	0.001	0.026
10%	LLM	0.001	0.001	0.000	0.000	0.002
10%	Human Validation	0.003	0.004	0.002	0.000	0.009
10%	Debiased	0.004	0.004	0.003	0.000	0.010
25%	LLM	0.001	0.001	0.001	0.000	0.002
25%	Human Validation	0.001	0.001	0.001	0.000	0.003
25%	Debiased	0.002	0.002	0.002	0.000	0.007
50%	LLM	0.001	0.001	0.000	0.000	0.002
50%	Human Validation	0.001	0.001	0.000	0.000	0.002
50%	Debiased	0.002	0.002	0.001	0.000	0.006

Table 38: MSE Summary Statistics of β_{Other} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	MSE	Mean	SD	Median	5%	95%
5%	LLM	0.000	0.000	0.000	0	0.000
5%	Human Validation	0.001	0.001	0.001	0	0.003
5%	Debiased	0.118	0.980	0.002	0	0.009
10%	LLM	0.000	0.000	0.000	0	0.000
10%	Human Validation	0.001	0.001	0.001	0	0.002
10%	Debiased	0.001	0.001	0.001	0	0.004
25%	LLM	0.000	0.000	0.000	0	0.000
25%	Human Validation	0.000	0.000	0.000	0	0.001
25%	Debiased	0.001	0.001	0.000	0	0.001
50%	LLM	0.000	0.000	0.000	0	0.000
50%	Human Validation	0.000	0.000	0.000	0	0.001
50%	Debiased	0.001	0.001	0.000	0	0.001

Table 39: MSE Summary Statistics of β_3 by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.001	0.001	0.001	0.000	0.003
GPT-3.5	5%	Human Validation	0.012	0.011	0.009	0.001	0.035
GPT-3.5	5%	Debiased	0.024	0.031	0.013	0.001	0.071
GPT-3.5	10%	LLM	0.002	0.001	0.001	0.000	0.004
GPT-3.5	10%	Human Validation	0.007	0.008	0.004	0.001	0.018
GPT-3.5	10%	Debiased	0.012	0.012	0.007	0.001	0.031
GPT-3.5	25%	LLM	0.002	0.002	0.001	0.000	0.005
GPT-3.5	25%	Human Validation	0.002	0.002	0.002	0.000	0.005
GPT-3.5	25%	Debiased	0.004	0.003	0.003	0.000	0.010
GPT-3.5	50%	LLM	0.001	0.001	0.001	0.000	0.003
GPT-3.5	50%	Human Validation	0.001	0.001	0.001	0.000	0.002
GPT-3.5	50%	Debiased	0.004	0.004	0.003	0.000	0.012
GPT-4o	5%	LLM	0.001	0.001	0.001	0.000	0.002
GPT-4o	5%	Human Validation	0.011	0.012	0.009	0.001	0.028
GPT-4o	5%	Debiased	0.008	0.012	0.005	0.001	0.029
GPT-4o	10%	LLM	0.001	0.001	0.001	0.000	0.002
GPT-4o	10%	Human Validation	0.007	0.006	0.006	0.001	0.020
GPT-4o	10%	Debiased	0.007	0.010	0.004	0.001	0.016
GPT-4o	25%	LLM	0.001	0.001	0.001	0.000	0.001
GPT-4o	25%	Human Validation	0.002	0.001	0.002	0.000	0.004
GPT-4o	25%	Debiased	0.002	0.002	0.002	0.000	0.005
GPT-4o	50%	LLM	0.001	0.001	0.000	0.000	0.001
GPT-4o	50%	Human Validation	0.001	0.001	0.001	0.000	0.002
GPT-4o	50%	Debiased	0.003	0.003	0.003	0.000	0.009

Table 40: MSE Summary Statistics of β_{14} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.002	0.001	0.001	0.000	0.004
GPT-3.5	5%	Human Validation	0.018	0.018	0.012	0.002	0.048
GPT-3.5	5%	Debiased	33.346	199.288	0.055	0.007	0.647
GPT-3.5	10%	LLM	0.002	0.001	0.002	0.000	0.004
GPT-3.5	10%	Human Validation	0.005	0.006	0.004	0.000	0.016
GPT-3.5	10%	Debiased	0.040	0.051	0.027	0.002	0.114
GPT-3.5	25%	LLM	0.002	0.002	0.001	0.000	0.006
GPT-3.5	25%	Human Validation	0.002	0.002	0.002	0.000	0.006
GPT-3.5	25%	Debiased	0.022	0.020	0.017	0.001	0.052
GPT-3.5	50%	LLM	0.002	0.001	0.002	0.000	0.004
GPT-3.5	50%	Human Validation	0.001	0.001	0.001	0.000	0.003
GPT-3.5	50%	Debiased	0.016	0.013	0.012	0.001	0.046
GPT-4o	5%	LLM	0.001	0.001	0.001	0.000	0.004
GPT-4o	5%	Human Validation	0.016	0.016	0.011	0.001	0.041
GPT-4o	5%	Debiased	0.086	0.130	0.034	0.004	0.416
GPT-4o	10%	LLM	0.001	0.001	0.001	0.000	0.003
GPT-4o	10%	Human Validation	0.009	0.008	0.006	0.001	0.026
GPT-4o	10%	Debiased	0.026	0.029	0.019	0.003	0.081
GPT-4o	25%	LLM	0.001	0.001	0.001	0.000	0.003
GPT-4o	25%	Human Validation	0.003	0.003	0.002	0.001	0.009
GPT-4o	25%	Debiased	0.011	0.010	0.007	0.002	0.027
GPT-4o	50%	LLM	0.001	0.001	0.001	0.000	0.002
GPT-4o	50%	Human Validation	0.002	0.002	0.001	0.000	0.004
GPT-4o	50%	Debiased	0.012	0.012	0.010	0.001	0.031

Table 41: MSE Summary Statistics of β_{15} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.001	0.000	0.000	0.000	0.001
GPT-3.5	5%	Human Validation	0.007	0.006	0.005	0.001	0.018
GPT-3.5	5%	Debiased	0.024	0.027	0.015	0.002	0.065
GPT-3.5	10%	LLM	0.001	0.001	0.000	0.000	0.003
GPT-3.5	10%	Human Validation	0.004	0.003	0.003	0.001	0.009
GPT-3.5	10%	Debiased	0.005	0.004	0.004	0.001	0.012
GPT-3.5	25%	LLM	0.001	0.001	0.000	0.000	0.002
GPT-3.5	25%	Human Validation	0.001	0.002	0.001	0.000	0.004
GPT-3.5	25%	Debiased	0.005	0.005	0.005	0.000	0.014
GPT-3.5	50%	LLM	0.001	0.001	0.000	0.000	0.002
GPT-3.5	50%	Human Validation	0.001	0.001	0.000	0.000	0.002
GPT-3.5	50%	Debiased	0.004	0.004	0.003	0.000	0.010
GPT-4o	5%	LLM	0.001	0.001	0.000	0.000	0.002
GPT-4o	5%	Human Validation	0.010	0.010	0.006	0.000	0.029
GPT-4o	5%	Debiased	0.014	0.013	0.011	0.000	0.035
GPT-4o	10%	LLM	0.001	0.001	0.001	0.000	0.002
GPT-4o	10%	Human Validation	0.003	0.002	0.002	0.000	0.008
GPT-4o	10%	Debiased	0.009	0.009	0.006	0.000	0.025
GPT-4o	25%	LLM	0.001	0.001	0.001	0.000	0.002
GPT-4o	25%	Human Validation	0.002	0.002	0.001	0.000	0.004
GPT-4o	25%	Debiased	0.004	0.004	0.003	0.000	0.011
GPT-4o	50%	LLM	0.001	0.001	0.001	0.000	0.002
GPT-4o	50%	Human Validation	0.001	0.001	0.001	0.000	0.002
GPT-4o	50%	Debiased	0.004	0.004	0.002	0.000	0.011

Table 42: MSE Summary Statistics of β_{19} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.001	0.001	0.001	0.000	0.002
GPT-3.5	5%	Human Validation	0.005	0.003	0.004	0.000	0.010
GPT-3.5	5%	Debiased	0.063	0.267	0.013	0.001	0.056
GPT-3.5	10%	LLM	0.001	0.001	0.000	0.000	0.001
GPT-3.5	10%	Human Validation	0.003	0.002	0.003	0.000	0.007
GPT-3.5	10%	Debiased	0.009	0.009	0.008	0.000	0.026
GPT-3.5	25%	LLM	0.001	0.000	0.001	0.000	0.001
GPT-3.5	25%	Human Validation	0.001	0.001	0.001	0.000	0.002
GPT-3.5	25%	Debiased	0.005	0.007	0.002	0.000	0.017
GPT-3.5	50%	LLM	0.001	0.001	0.000	0.000	0.002
GPT-3.5	50%	Human Validation	0.000	0.000	0.000	0.000	0.001
GPT-3.5	50%	Debiased	0.003	0.003	0.003	0.000	0.009
GPT-4o	5%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-4o	5%	Human Validation	0.006	0.005	0.005	0.000	0.015
GPT-4o	5%	Debiased	0.009	0.007	0.007	0.000	0.019
GPT-4o	10%	LLM	0.000	0.000	0.000	0.000	0.002
GPT-4o	10%	Human Validation	0.003	0.003	0.003	0.000	0.009
GPT-4o	10%	Debiased	0.007	0.006	0.005	0.001	0.018
GPT-4o	25%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-4o	25%	Human Validation	0.001	0.001	0.001	0.000	0.003
GPT-4o	25%	Debiased	0.003	0.003	0.002	0.000	0.008
GPT-4o	50%	LLM	0.000	0.000	0.000	0.000	0.001
GPT-4o	50%	Human Validation	0.001	0.000	0.000	0.000	0.001
GPT-4o	50%	Debiased	0.002	0.002	0.002	0.001	0.005

Table 43: MSE Summary Statistics of β_{20} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.001	0.001	0.001	0.000	0.002
GPT-3.5	5%	Human Validation	0.007	0.007	0.006	0.001	0.019
GPT-3.5	5%	Debiased	0.012	0.010	0.011	0.000	0.029
GPT-3.5	10%	LLM	0.001	0.000	0.000	0.000	0.001
GPT-3.5	10%	Human Validation	0.004	0.004	0.003	0.000	0.011
GPT-3.5	10%	Debiased	0.004	0.004	0.003	0.000	0.008
GPT-3.5	25%	LLM	0.001	0.001	0.000	0.000	0.001
GPT-3.5	25%	Human Validation	0.001	0.001	0.001	0.000	0.002
GPT-3.5	25%	Debiased	0.002	0.002	0.002	0.000	0.005
GPT-3.5	50%	LLM	0.001	0.000	0.000	0.000	0.001
GPT-3.5	50%	Human Validation	0.001	0.001	0.000	0.000	0.002
GPT-3.5	50%	Debiased	0.002	0.002	0.001	0.000	0.005
GPT-4o	5%	LLM	0.001	0.001	0.000	0.000	0.002
GPT-4o	5%	Human Validation	0.006	0.005	0.005	0.000	0.016
GPT-4o	5%	Debiased	0.009	0.007	0.009	0.001	0.025
GPT-4o	10%	LLM	0.001	0.001	0.001	0.000	0.002
GPT-4o	10%	Human Validation	0.003	0.003	0.002	0.000	0.006
GPT-4o	10%	Debiased	0.004	0.003	0.002	0.000	0.011
GPT-4o	25%	LLM	0.001	0.001	0.001	0.000	0.002
GPT-4o	25%	Human Validation	0.001	0.001	0.001	0.000	0.003
GPT-4o	25%	Debiased	0.002	0.002	0.001	0.000	0.007
GPT-4o	50%	LLM	0.001	0.001	0.000	0.000	0.002
GPT-4o	50%	Human Validation	0.001	0.001	0.001	0.000	0.002
GPT-4o	50%	Debiased	0.002	0.002	0.001	0.000	0.006

Table 44: MSE Summary Statistics of β_{Other} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^{\top}\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.000	0.000	0.000	0	0.000
GPT-3.5	5%	Human Validation	0.001	0.001	0.001	0	0.003
GPT-3.5	5%	Debiased	0.235	1.386	0.003	0	0.012
GPT-3.5	10%	LLM	0.000	0.000	0.000	0	0.000
GPT-3.5	10%	Human Validation	0.001	0.001	0.001	0	0.002
GPT-3.5	10%	Debiased	0.002	0.002	0.001	0	0.006
GPT-3.5	25%	LLM	0.000	0.000	0.000	0	0.000
GPT-3.5	25%	Human Validation	0.000	0.000	0.000	0	0.001
GPT-3.5	25%	Debiased	0.001	0.001	0.001	0	0.002
GPT-3.5	50%	LLM	0.000	0.000	0.000	0	0.000
GPT-3.5	50%	Human Validation	0.000	0.000	0.000	0	0.000
GPT-3.5	50%	Debiased	0.001	0.001	0.000	0	0.002
GPT-4o	5%	LLM	0.000	0.000	0.000	0	0.000
GPT-4o	5%	Human Validation	0.001	0.001	0.001	0	0.003
GPT-4o	5%	Debiased	0.002	0.002	0.001	0	0.007
GPT-4o	10%	LLM	0.000	0.000	0.000	0	0.000
GPT-4o	10%	Human Validation	0.001	0.001	0.001	0	0.002
GPT-4o	10%	Debiased	0.001	0.001	0.001	0	0.002
GPT-4o	25%	LLM	0.000	0.000	0.000	0	0.000
GPT-4o	25%	Human Validation	0.000	0.000	0.000	0	0.001
GPT-4o	25%	Debiased	0.000	0.001	0.000	0	0.001
GPT-4o	50%	LLM	0.000	0.000	0.000	0	0.000
GPT-4o	50%	Human Validation	0.000	0.000	0.000	0	0.001
GPT-4o	50%	Debiased	0.001	0.000	0.000	0	0.001

Table 45: Coverage Summary Statistics of β_3 by Proportion of Validation Samples. Proxy on the RHS: $V = Y^{\top}\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.868	0.212	1.000	0.5	1
5%	Human Validation	0.914	0.179	1.000	0.5	1
5%	Debiased	0.797	0.254	1.000	0.5	1
10%	LLM	0.848	0.223	1.000	0.5	1
10%	Human Validation	0.924	0.172	1.000	0.5	1
10%	Debiased	0.709	0.306	0.714	0.0	1
25%	LLM	0.862	0.231	1.000	0.5	1
25%	Human Validation	0.952	0.138	1.000	0.5	1
25%	Debiased	0.820	0.250	1.000	0.5	1
50%	LLM	0.876	0.224	1.000	0.5	1
50%	Human Validation	0.960	0.127	1.000	0.5	1
50%	Debiased	0.799	0.255	1.000	0.5	1

Table 46: Coverage Summary Statistics of β_{14} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.680	0.346	0.728	0.000	1
5%	Human Validation	0.901	0.209	1.000	0.500	1
5%	Debiased	0.819	0.264	1.000	0.500	1
10%	LLM	0.700	0.350	0.916	0.000	1
10%	Human Validation	0.916	0.179	1.000	0.500	1
10%	Debiased	0.827	0.248	1.000	0.500	1
25%	LLM	0.700	0.351	0.907	0.000	1
25%	Human Validation	0.959	0.127	1.000	0.500	1
25%	Debiased	0.751	0.274	0.950	0.500	1
50%	LLM	0.659	0.362	0.651	0.000	1
50%	Human Validation	0.959	0.128	1.000	0.500	1
50%	Debiased	0.779	0.297	1.000	0.275	1

Table 47: Coverage Summary Statistics of β_{15} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.929	0.166	1.000	0.500	1
5%	Human Validation	0.923	0.172	1.000	0.500	1
5%	Debiased	0.764	0.331	1.000	0.000	1
10%	LLM	0.900	0.192	1.000	0.500	1
10%	Human Validation	0.966	0.115	1.000	0.737	1
10%	Debiased	0.834	0.260	1.000	0.500	1
25%	LLM	0.921	0.192	1.000	0.500	1
25%	Human Validation	0.952	0.138	1.000	0.500	1
25%	Debiased	0.800	0.294	1.000	0.275	1
50%	LLM	0.914	0.180	1.000	0.500	1
50%	Human Validation	0.959	0.127	1.000	0.500	1
50%	Debiased	0.751	0.321	0.959	0.000	1

Table 48: Coverage Summary Statistics of β_{19} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.904	0.221	1.000	0.500	1
5%	Human Validation	0.952	0.138	1.000	0.500	1
5%	Debiased	0.806	0.267	1.000	0.500	1
10%	LLM	0.939	0.151	1.000	0.500	1
10%	Human Validation	0.923	0.172	1.000	0.500	1
10%	Debiased	0.772	0.273	0.951	0.500	1
25%	LLM	0.919	0.193	1.000	0.500	1
25%	Human Validation	0.966	0.115	1.000	0.741	1
25%	Debiased	0.820	0.264	1.000	0.500	1
50%	LLM	0.911	0.216	1.000	0.500	1
50%	Human Validation	0.967	0.115	1.000	0.745	1
50%	Debiased	0.779	0.308	1.000	0.000	1

Table 49: Coverage Summary Statistics of β_{20} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.893	0.197	1.00	0.5	1
5%	Human Validation	0.945	0.148	1.00	0.5	1
5%	Debiased	0.751	0.310	0.95	0.0	1
10%	LLM	0.908	0.186	1.00	0.5	1
10%	Human Validation	0.952	0.138	1.00	0.5	1
10%	Debiased	0.813	0.252	1.00	0.5	1
25%	LLM	0.880	0.223	1.00	0.5	1
25%	Human Validation	0.952	0.138	1.00	0.5	1
25%	Debiased	0.786	0.284	1.00	0.5	1
50%	LLM	0.909	0.204	1.00	0.5	1
50%	Human Validation	0.960	0.127	1.00	0.5	1
50%	Debiased	0.821	0.264	1.00	0.5	1

Table 50: Coverage Summary Statistics of β_{Other} by Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Validation Proportion	Coverage	Mean	SD	Median	5%	95%
5%	LLM	0.830	0.296	1.000	0.000	1
5%	Human Validation	0.966	0.115	1.000	0.741	1
5%	Debiased	0.814	0.253	1.000	0.500	1
10%	LLM	0.865	0.259	1.000	0.500	1
10%	Human Validation	0.946	0.148	1.000	0.500	1
10%	Debiased	0.786	0.272	0.981	0.500	1
25%	LLM	0.816	0.288	1.000	0.275	1
25%	Human Validation	0.918	0.198	1.000	0.500	1
25%	Debiased	0.834	0.273	1.000	0.500	1
50%	LLM	0.837	0.256	1.000	0.500	1
50%	Human Validation	0.960	0.127	1.000	0.500	1
50%	Debiased	0.793	0.270	1.000	0.500	1

Table 51: Coverage Summary Statistics of β_3 by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.829	0.230	1.000	0.500	1
GPT-3.5	5%	Human Validation	0.921	0.174	1.000	0.500	1
GPT-3.5	5%	Debiased	0.811	0.267	1.000	0.500	1
GPT-3.5	10%	LLM	0.774	0.243	0.942	0.500	1
GPT-3.5	10%	Human Validation	0.896	0.198	1.000	0.500	1
GPT-3.5	10%	Debiased	0.687	0.293	0.500	0.375	1
GPT-3.5	25%	LLM	0.802	0.266	1.000	0.500	1
GPT-3.5	25%	Human Validation	0.924	0.174	1.000	0.500	1
GPT-3.5	25%	Debiased	0.813	0.240	1.000	0.500	1
GPT-3.5	50%	LLM	0.816	0.264	1.000	0.500	1
GPT-3.5	50%	Human Validation	0.953	0.139	1.000	0.500	1
GPT-3.5	50%	Debiased	0.785	0.273	0.973	0.500	1
GPT-4o	5%	LLM	0.908	0.187	1.000	0.500	1
GPT-4o	5%	Human Validation	0.907	0.187	1.000	0.500	1
GPT-4o	5%	Debiased	0.783	0.244	0.929	0.500	1
GPT-4o	10%	LLM	0.922	0.174	1.000	0.500	1
GPT-4o	10%	Human Validation	0.951	0.139	1.000	0.500	1
GPT-4o	10%	Debiased	0.730	0.322	0.942	0.000	1
GPT-4o	25%	LLM	0.922	0.174	1.000	0.500	1
GPT-4o	25%	Human Validation	0.980	0.084	1.000	0.945	1
GPT-4o	25%	Debiased	0.828	0.264	1.000	0.500	1
GPT-4o	50%	LLM	0.936	0.158	1.000	0.500	1
GPT-4o	50%	Human Validation	0.966	0.116	1.000	0.831	1
GPT-4o	50%	Debiased	0.814	0.240	1.000	0.500	1

Table 52: Coverage Summary Statistics of β_{14} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.598	0.375	0.500	0.000	1
GPT-3.5	5%	Human Validation	0.880	0.239	1.000	0.500	1
GPT-3.5	5%	Debiased	0.855	0.280	1.000	0.375	1
GPT-3.5	10%	LLM	0.598	0.357	0.500	0.000	1
GPT-3.5	10%	Human Validation	0.923	0.174	1.000	0.500	1
GPT-3.5	10%	Debiased	0.826	0.264	1.000	0.500	1
GPT-3.5	25%	LLM	0.612	0.383	0.500	0.000	1
GPT-3.5	25%	Human Validation	0.980	0.084	1.000	0.942	1
GPT-3.5	25%	Debiased	0.717	0.274	0.500	0.500	1
GPT-3.5	50%	LLM	0.529	0.379	0.500	0.000	1
GPT-3.5	50%	Human Validation	0.980	0.084	1.000	0.943	1
GPT-3.5	50%	Debiased	0.800	0.296	1.000	0.375	1
GPT-4o	5%	LLM	0.761	0.298	1.000	0.360	1
GPT-4o	5%	Human Validation	0.922	0.174	1.000	0.500	1
GPT-4o	5%	Debiased	0.784	0.244	0.940	0.500	1
GPT-4o	10%	LLM	0.803	0.317	1.000	0.000	1
GPT-4o	10%	Human Validation	0.909	0.187	1.000	0.500	1
GPT-4o	10%	Debiased	0.827	0.235	1.000	0.500	1
GPT-4o	25%	LLM	0.788	0.296	1.000	0.353	1
GPT-4o	25%	Human Validation	0.939	0.158	1.000	0.500	1
GPT-4o	25%	Debiased	0.786	0.273	0.976	0.500	1
GPT-4o	50%	LLM	0.788	0.296	1.000	0.352	1
GPT-4o	50%	Human Validation	0.939	0.158	1.000	0.500	1
GPT-4o	50%	Debiased	0.758	0.300	0.955	0.375	1

Table 53: Coverage Summary Statistics of β_{15} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.920	0.175	1.000	0.500	1
GPT-3.5	5%	Human Validation	0.937	0.158	1.000	0.500	1
GPT-3.5	5%	Debiased	0.798	0.319	1.000	0.000	1
GPT-3.5	10%	LLM	0.850	0.223	1.000	0.500	1
GPT-3.5	10%	Human Validation	0.952	0.139	1.000	0.500	1
GPT-3.5	10%	Debiased	0.855	0.254	1.000	0.500	1
GPT-3.5	25%	LLM	0.905	0.223	1.000	0.500	1
GPT-3.5	25%	Human Validation	0.938	0.158	1.000	0.500	1
GPT-3.5	25%	Debiased	0.786	0.298	1.000	0.375	1
GPT-3.5	50%	LLM	0.905	0.188	1.000	0.500	1
GPT-3.5	50%	Human Validation	0.952	0.139	1.000	0.500	1
GPT-3.5	50%	Debiased	0.800	0.271	1.000	0.500	1
GPT-4o	5%	LLM	0.937	0.158	1.000	0.500	1
GPT-4o	5%	Human Validation	0.909	0.187	1.000	0.500	1
GPT-4o	5%	Debiased	0.729	0.343	0.939	0.000	1
GPT-4o	10%	LLM	0.951	0.140	1.000	0.500	1
GPT-4o	10%	Human Validation	0.979	0.084	1.000	0.937	1
GPT-4o	10%	Debiased	0.813	0.268	1.000	0.500	1
GPT-4o	25%	LLM	0.937	0.158	1.000	0.500	1
GPT-4o	25%	Human Validation	0.966	0.116	1.000	0.824	1
GPT-4o	25%	Debiased	0.814	0.293	1.000	0.375	1
GPT-4o	50%	LLM	0.923	0.174	1.000	0.500	1
GPT-4o	50%	Human Validation	0.966	0.116	1.000	0.830	1
GPT-4o	50%	Debiased	0.702	0.361	0.942	0.000	1

Table 54: Coverage Summary Statistics of β_{19} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^{\top}\beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.873	0.240	1.000	0.500	1
GPT-3.5	5%	Human Validation	0.965	0.116	1.000	0.825	1
GPT-3.5	5%	Debiased	0.771	0.299	0.972	0.375	1
GPT-3.5	10%	LLM	0.928	0.162	1.000	0.500	1
GPT-3.5	10%	Human Validation	0.909	0.187	1.000	0.500	1
GPT-3.5	10%	Debiased	0.786	0.298	1.000	0.375	1
GPT-3.5	25%	LLM	0.915	0.176	1.000	0.500	1
GPT-3.5	25%	Human Validation	0.966	0.116	1.000	0.829	1
GPT-3.5	25%	Debiased	0.827	0.264	1.000	0.500	1
GPT-3.5	50%	LLM	0.858	0.275	1.000	0.375	1
GPT-3.5	50%	Human Validation	0.994	0.016	1.000	0.950	1
GPT-3.5	50%	Debiased	0.813	0.268	1.000	0.500	1
GPT-4o	5%	LLM	0.936	0.198	1.000	0.500	1
GPT-4o	5%	Human Validation	0.938	0.158	1.000	0.500	1
GPT-4o	5%	Debiased	0.841	0.230	1.000	0.500	1
GPT-4o	10%	LLM	0.950	0.140	1.000	0.500	1
GPT-4o	10%	Human Validation	0.937	0.158	1.000	0.500	1
GPT-4o	10%	Debiased	0.758	0.248	0.944	0.500	1
GPT-4o	25%	LLM	0.922	0.211	1.000	0.500	1
GPT-4o	25%	Human Validation	0.966	0.116	1.000	0.833	1
GPT-4o	25%	Debiased	0.813	0.268	1.000	0.500	1
GPT-4o	50%	LLM	0.963	0.117	1.000	0.790	1
GPT-4o	50%	Human Validation	0.940	0.158	1.000	0.500	1
GPT-4o	50%	Debiased	0.744	0.345	0.977	0.000	1

Table 55: Coverage Summary Statistics of β_{20} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.881	0.208	1.000	0.500	1
GPT-3.5	5%	Human Validation	0.924	0.174	1.000	0.500	1
GPT-3.5	5%	Debiased	0.800	0.243	0.978	0.500	1
GPT-3.5	10%	LLM	0.923	0.174	1.000	0.500	1
GPT-3.5	10%	Human Validation	0.952	0.139	1.000	0.500	1
GPT-3.5	10%	Debiased	0.827	0.235	1.000	0.500	1
GPT-3.5	25%	LLM	0.909	0.222	1.000	0.500	1
GPT-3.5	25%	Human Validation	0.980	0.084	1.000	0.937	1
GPT-3.5	25%	Debiased	0.786	0.298	1.000	0.375	1
GPT-3.5	50%	LLM	0.938	0.158	1.000	0.500	1
GPT-3.5	50%	Human Validation	0.953	0.139	1.000	0.500	1
GPT-3.5	50%	Debiased	0.856	0.254	1.000	0.500	1
GPT-4o	5%	LLM	0.906	0.187	1.000	0.500	1
GPT-4o	5%	Human Validation	0.966	0.116	1.000	0.824	1
GPT-4o	5%	Debiased	0.702	0.361	0.940	0.000	1
GPT-4o	10%	LLM	0.894	0.199	1.000	0.500	1
GPT-4o	10%	Human Validation	0.952	0.139	1.000	0.500	1
GPT-4o	10%	Debiased	0.800	0.271	1.000	0.500	1
GPT-4o	25%	LLM	0.851	0.223	1.000	0.500	1
GPT-4o	25%	Human Validation	0.925	0.174	1.000	0.500	1
GPT-4o	25%	Debiased	0.786	0.273	0.978	0.500	1
GPT-4o	50%	LLM	0.880	0.240	1.000	0.500	1
GPT-4o	50%	Human Validation	0.967	0.116	1.000	0.835	1
GPT-4o	50%	Debiased	0.786	0.273	0.980	0.500	1

Table 56: Coverage Summary Statistics of β_{Other} by Model and Proportion of Validation Samples. Proxy on the RHS: $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

Model	Validation Proportion	MSE	Mean	SD	Median	5%	95%
GPT-3.5	5%	LLM	0.726	0.362	1.000	0.000	1
GPT-3.5	5%	Human Validation	0.980	0.084	1.000	0.945	1
GPT-3.5	5%	Debiased	0.828	0.264	1.000	0.500	1
GPT-3.5	10%	LLM	0.768	0.320	1.000	0.000	1
GPT-3.5	10%	Human Validation	0.939	0.158	1.000	0.500	1
GPT-3.5	10%	Debiased	0.800	0.271	1.000	0.500	1
GPT-3.5	25%	LLM	0.754	0.342	1.000	0.000	1
GPT-3.5	25%	Human Validation	0.911	0.222	1.000	0.500	1
GPT-3.5	25%	Debiased	0.869	0.247	1.000	0.500	1
GPT-3.5	50%	LLM	0.810	0.266	1.000	0.500	1
GPT-3.5	50%	Human Validation	0.994	0.016	1.000	0.952	1
GPT-3.5	50%	Debiased	0.841	0.230	1.000	0.500	1
GPT-4o	5%	LLM	0.933	0.159	1.000	0.500	1
GPT-4o	5%	Human Validation	0.952	0.139	1.000	0.500	1
GPT-4o	5%	Debiased	0.800	0.243	0.984	0.500	1
GPT-4o	10%	LLM	0.961	0.119	1.000	0.765	1
GPT-4o	10%	Human Validation	0.952	0.139	1.000	0.500	1
GPT-4o	10%	Debiased	0.773	0.275	0.959	0.500	1
GPT-4o	25%	LLM	0.878	0.208	1.000	0.500	1
GPT-4o	25%	Human Validation	0.925	0.174	1.000	0.500	1
GPT-4o	25%	Debiased	0.800	0.296	1.000	0.375	1
GPT-4o	50%	LLM	0.864	0.247	1.000	0.500	1
GPT-4o	50%	Human Validation	0.926	0.174	1.000	0.500	1
GPT-4o	50%	Debiased	0.745	0.301	0.952	0.375	1